

Static probabilistic timing analysis of worst-case execution time for random replacement caches

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Research team: ALF

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Hard real-time systems

Introduction

Required background

Proposed method

Conclusion

Real-time systems

Estimation techniques

State of the art



Soft real-time systems

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Real-time programs

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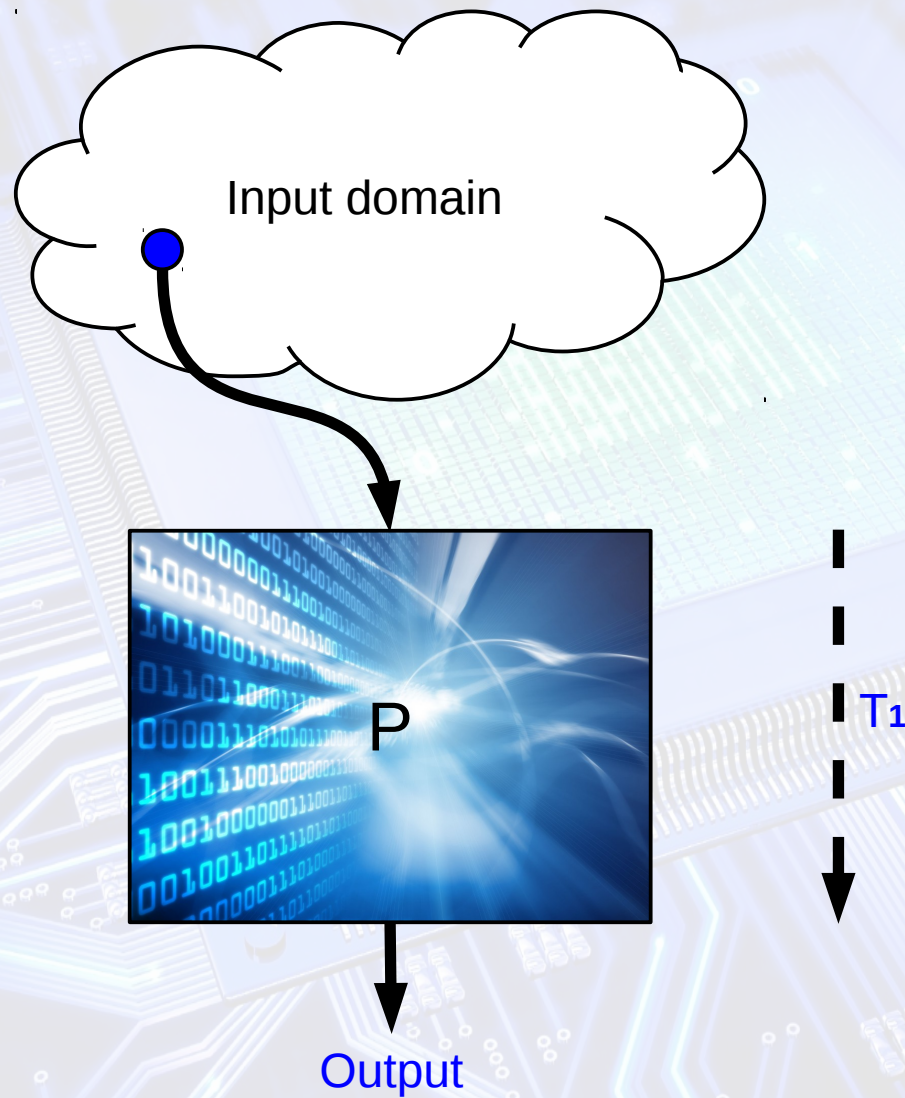
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Real-time programs

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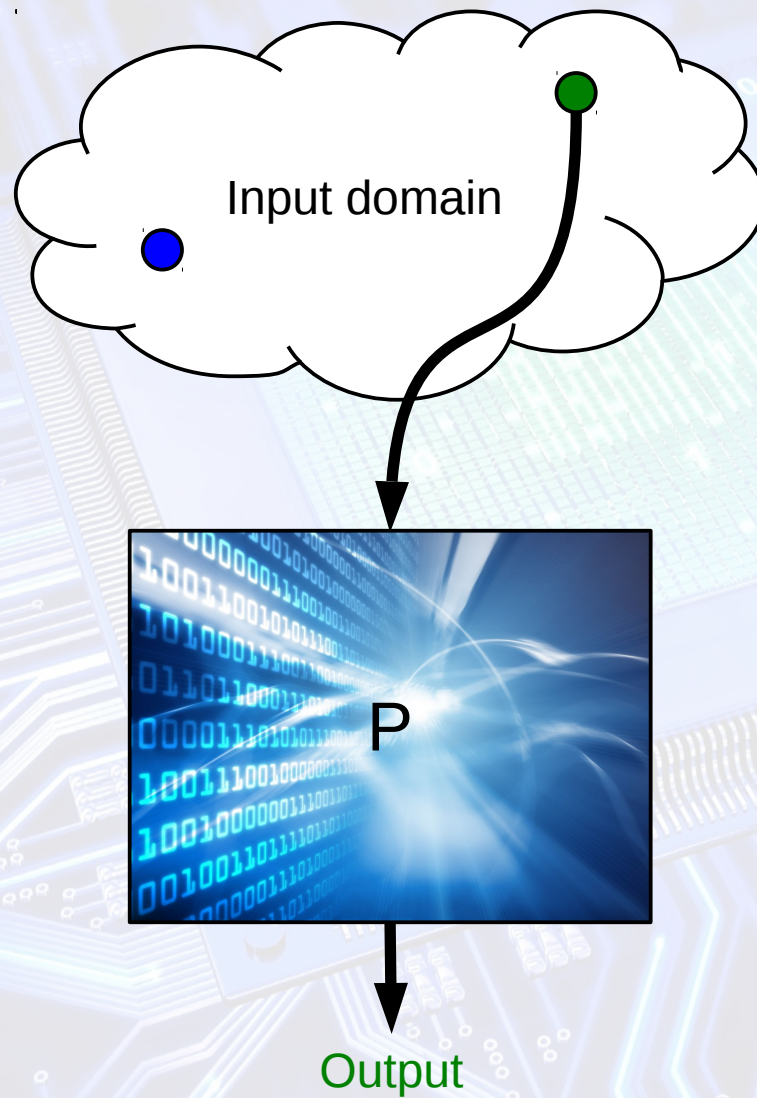
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Real-time programs

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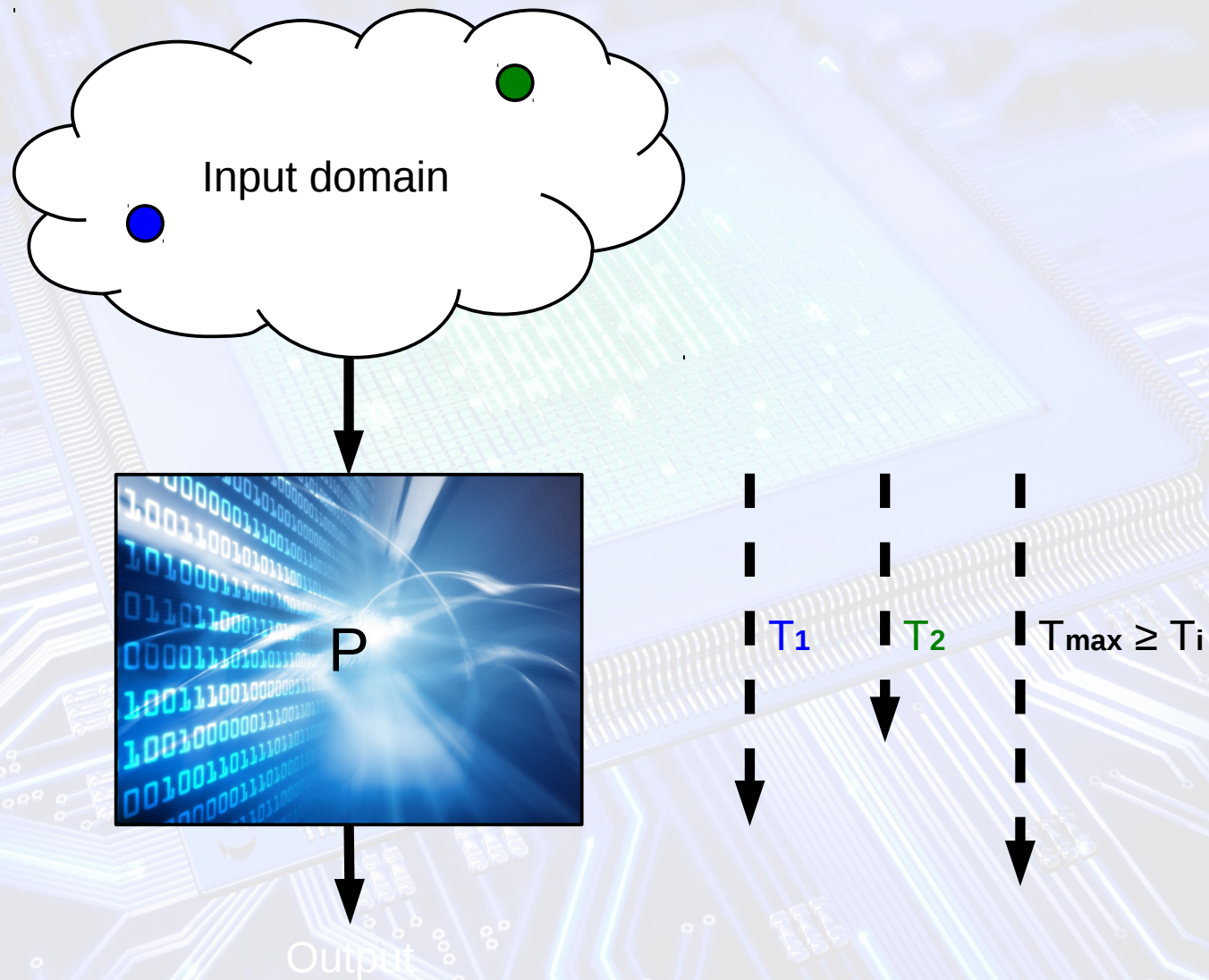
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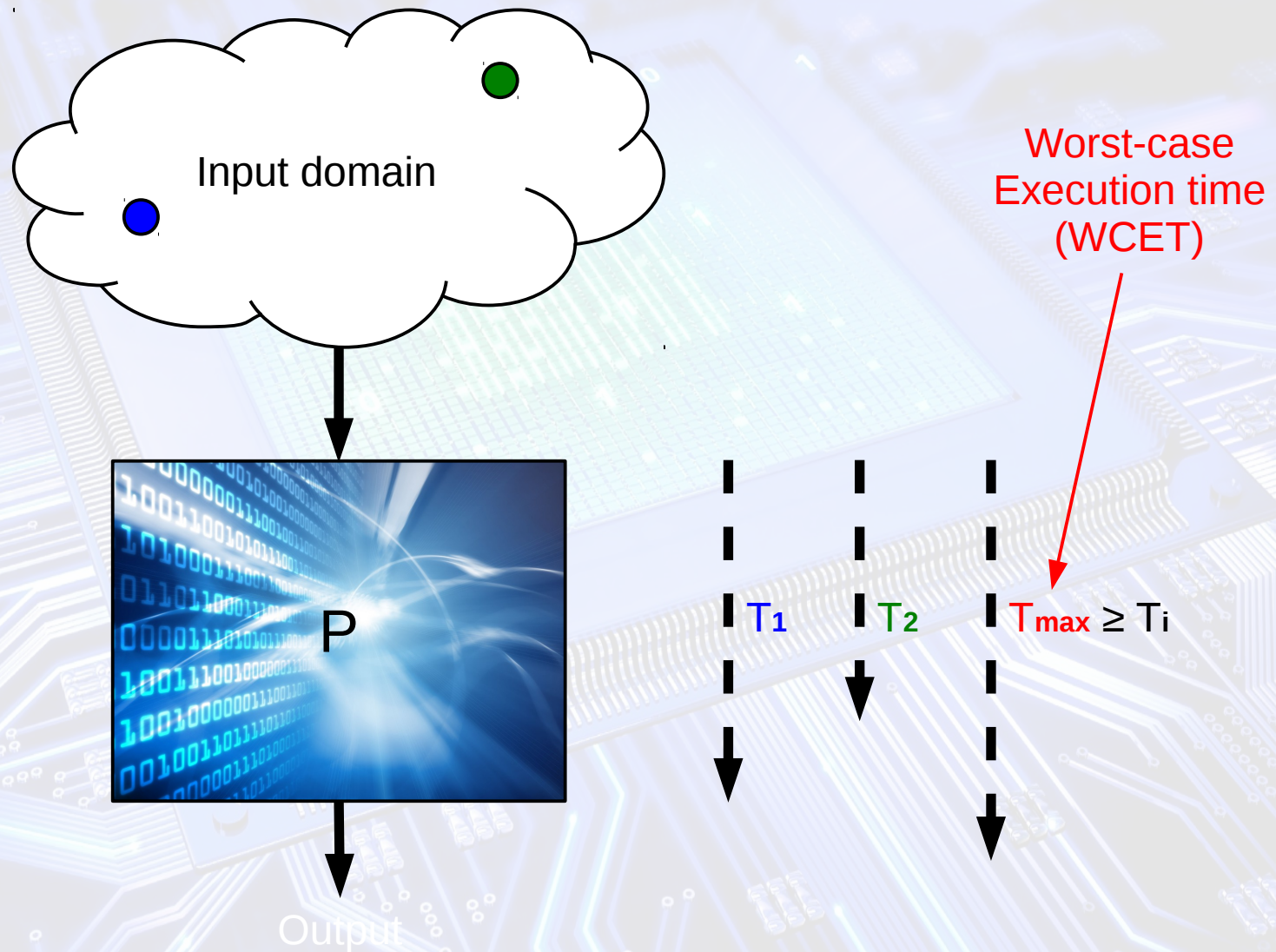
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Scheduling of real-time programs

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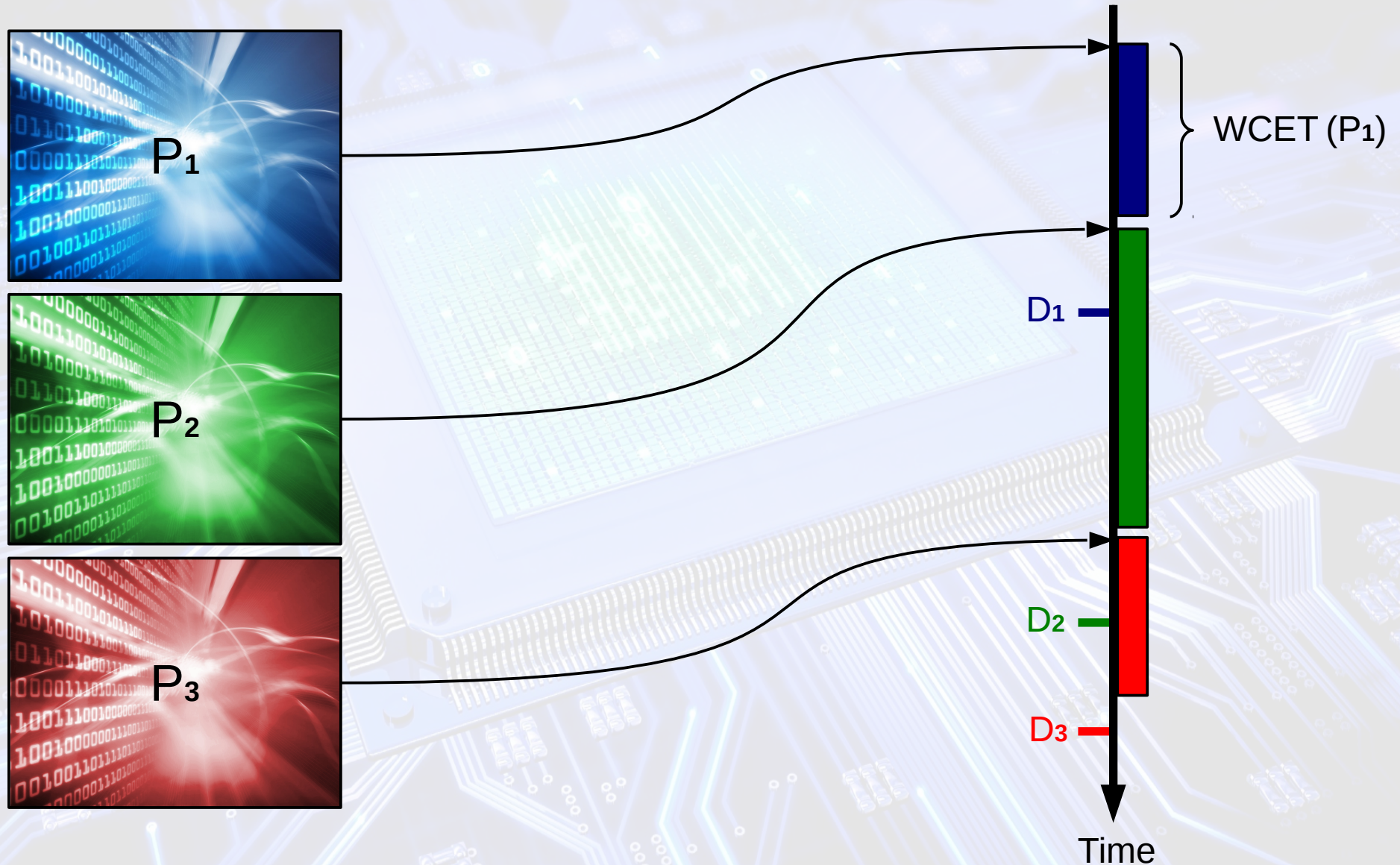
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WCET estimation

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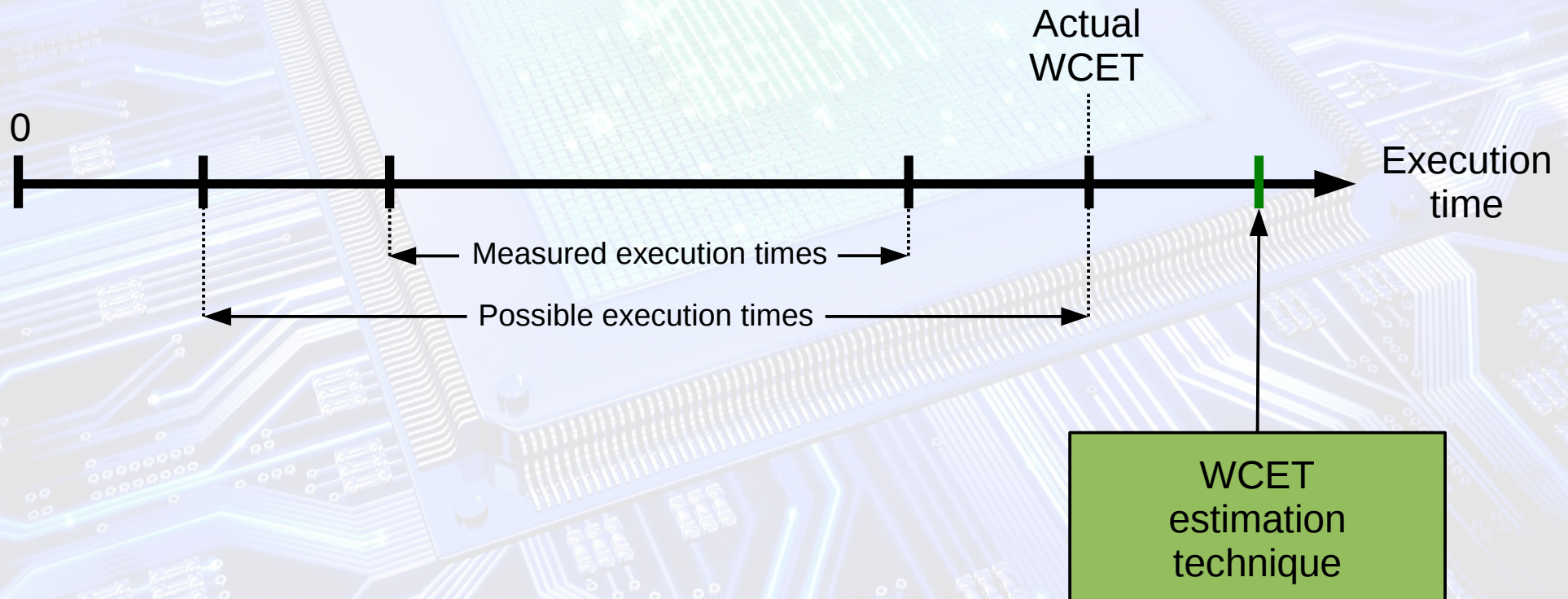
State of the art

Safety

Estimate greater than the actual WCET

Tightness

As close as possible to the actual WCET



WCET estimation

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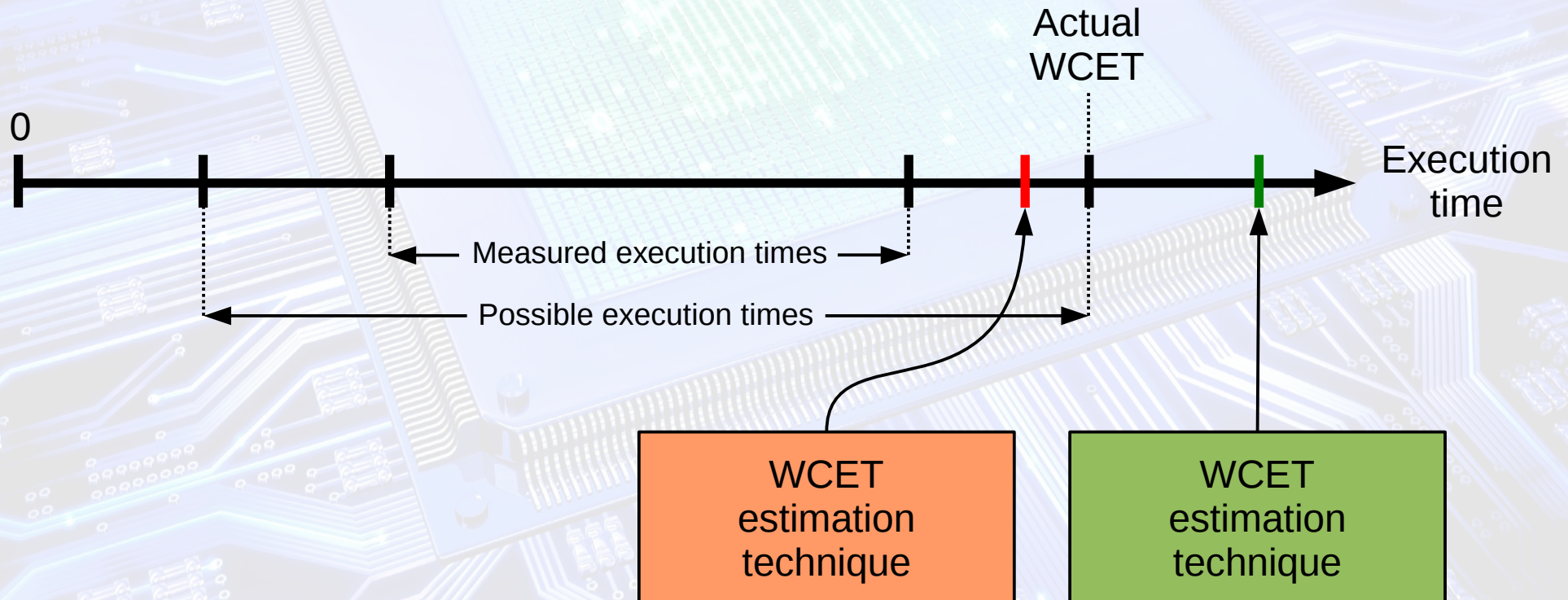
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WCET estimation

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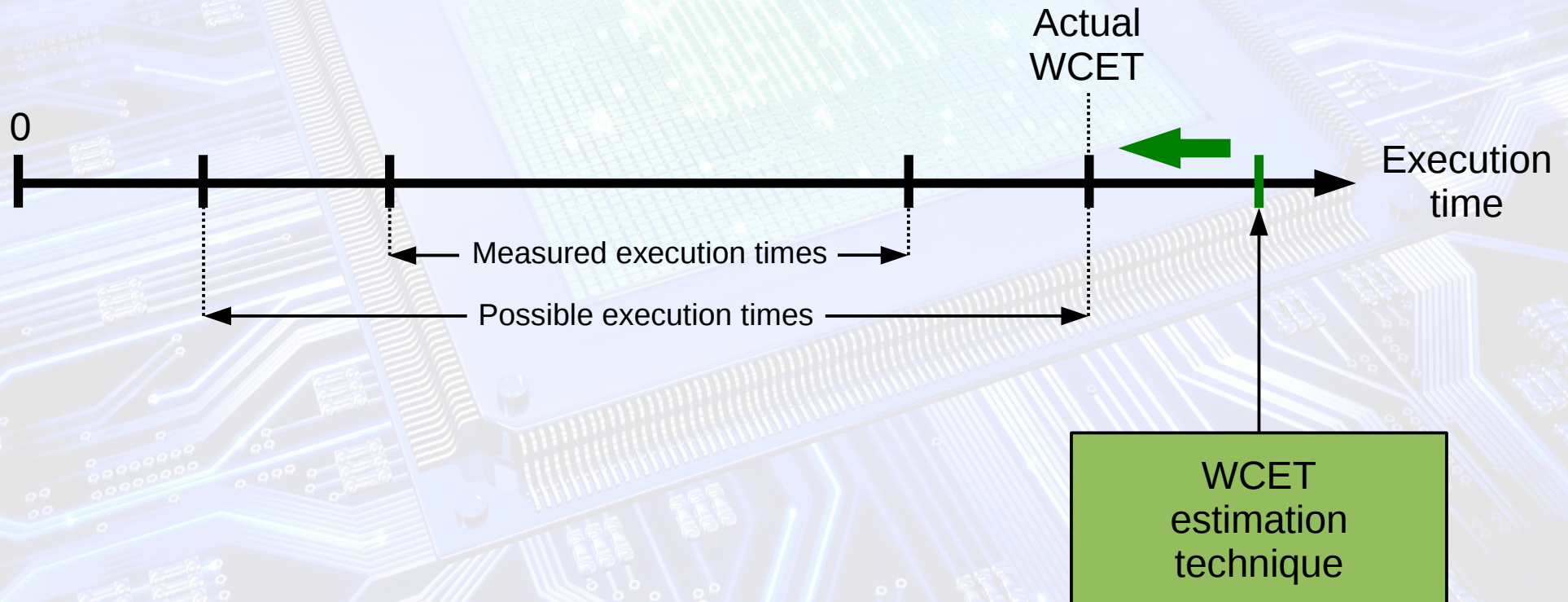
State of the art

Safety

Estimate greater than the actual WCET

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As close as possible to the actual WCET



Deterministic Vs. probabilistic

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Deterministic techniques



**Deterministic
WCET estimation
technique**

Single WCET estimate

Deterministic Vs. probabilistic

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Deterministic techniques



**Deterministic
WCET estimation
technique**

Single WCET estimate

Probabilistic techniques



**Probabilistic
WCET estimation
technique**

**Execution time profile
(timing values + probabilities)**

Deterministic Vs. probabilistic

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Deterministic techniques



**Deterministic
WCET estimation
technique**

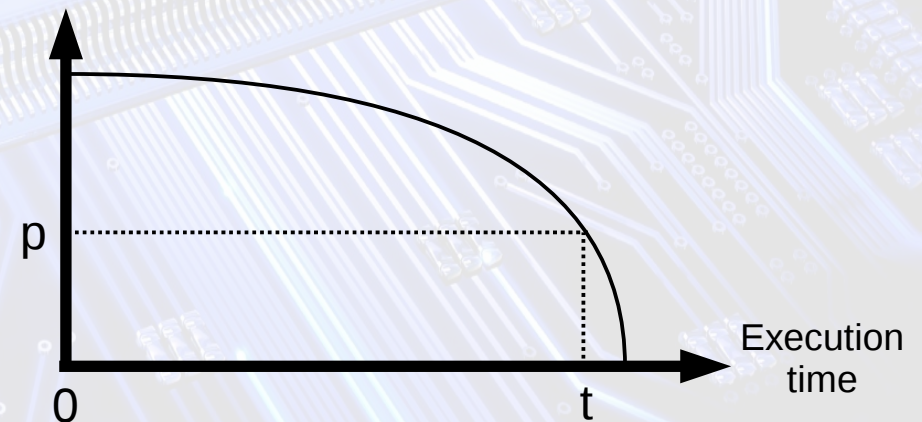
Single WCET estimate

Probabilistic techniques

**Probabilistic
WCET estimation
technique**

**Execution time profile
(timing values + probabilities)**

Probability of
exceedance



Execution
time

Static Vs. Measurement-based

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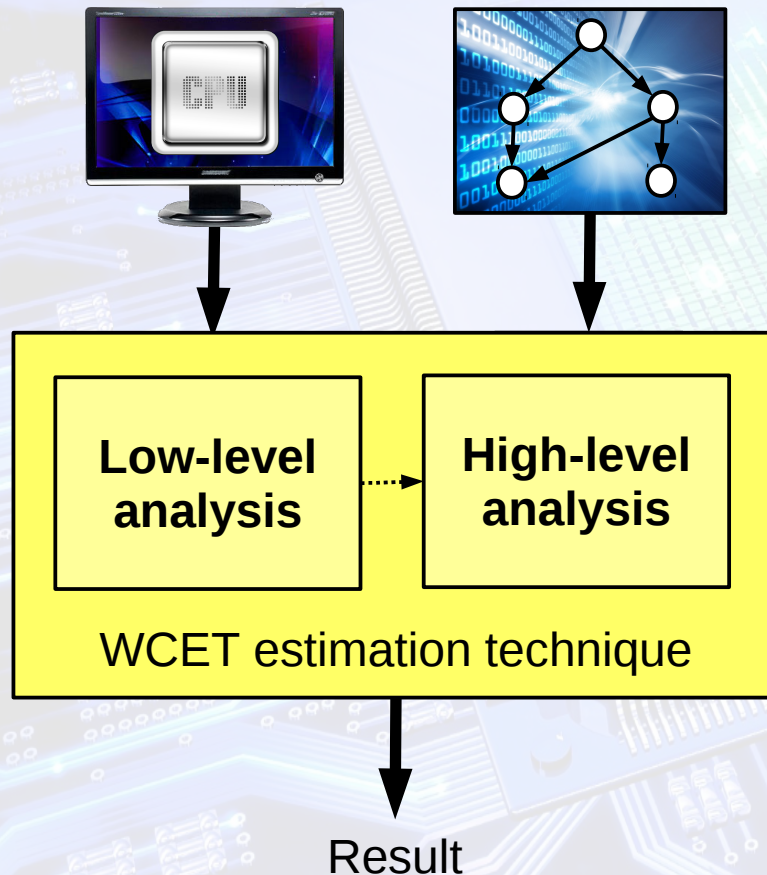
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Static Vs. Measurement-based

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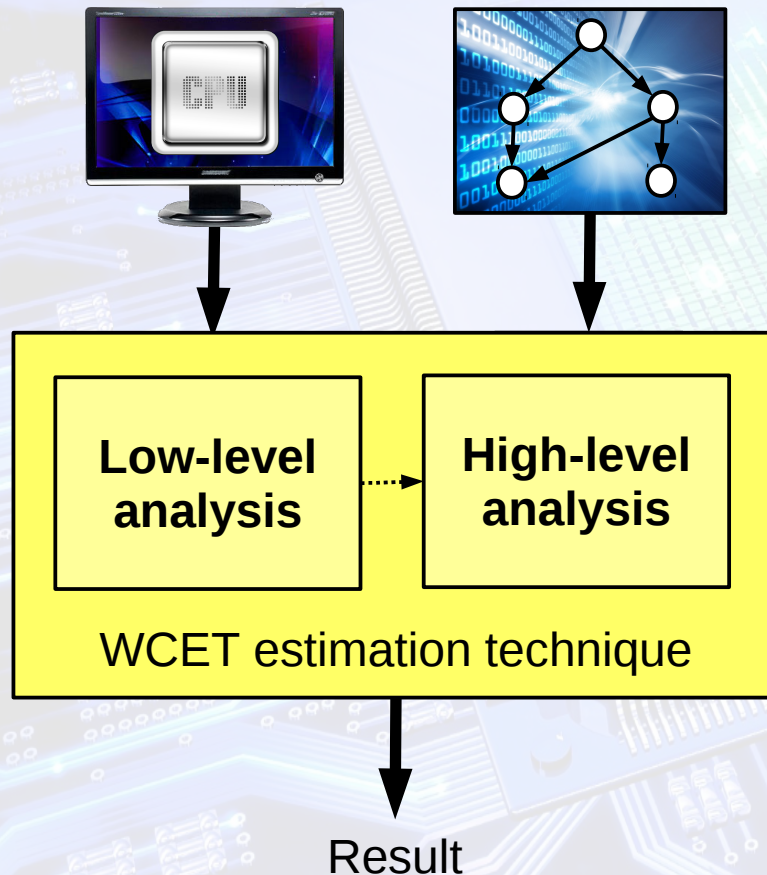
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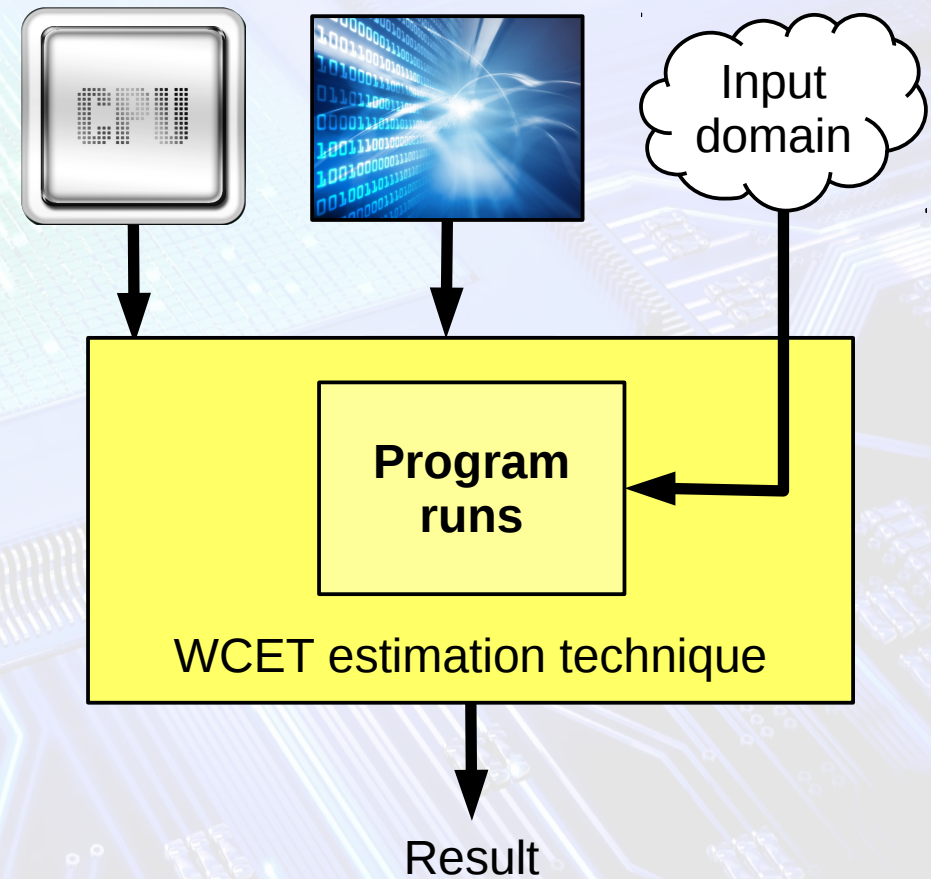
Estimation techniques

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Static techniques



Measurement-based techniques



Overview of the methods

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		Deterministic	Probabilistic
Static methods	High-level	[Pus1989] [Ste1995]	[Dav2004]
	Low-level	[Fer1999]	[Alt2014]
Measurement-based methods		[Wil2005] [Wil2008]	[Ber2002] [Cuc2012]

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- [Alt2014] S. Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis" (2014)
[Ber2002] G. Bernat, A. Colin, S.M. Petters – "WCET Analysis of Probabilistic Hard Real-Time Systems" (2002)
[Cuc2012] L. Cucu-Grosjean *et. al* – "Measurement-Based Probabilistic Timing Analysis for Multi-path Programs" (2012)
[Dav2004] L. David, I. Puaut – "Static Determination of Probabilistic Execution Times" (2004)
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[Pus1989] P. Puschner, C. Koza – "Calculating the Maximum Execution Time of Real-Time Programs" (1989)
[Ste1995] Y.T. Steven Li, S. Malik – "Performance Analysis of Embedded Software Using Implicit Path Enumeration" (1995)
[Wil2008] R. Wilhelm *et. al* – "The Worst-case Execution-time Problem – Overview of Methods and Survey of Tools" (2008)
[Wil2005] N. Williams *et. al* – "PathCrawler: automatic generation of path tests by combining static and dynamic analysis" (2005)

Overview of the methods

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Static probabilistic reference method [Alt2014]

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Real-time systems

Estimation techniques

State of the art

- ▶ Probabilistic
- ▶ Static
- ▶ Low-level
- ▶ Single-path

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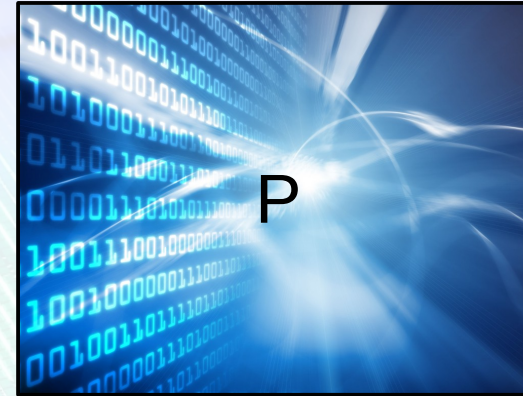
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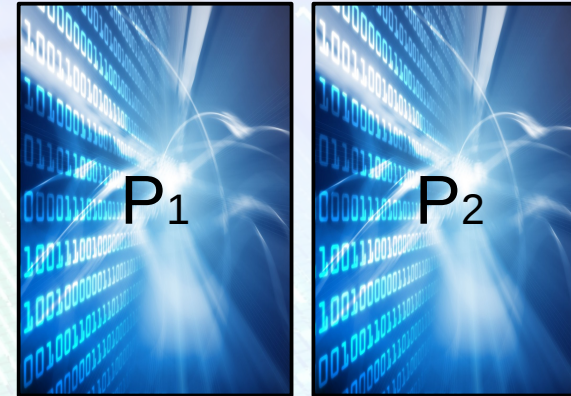
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Real-time systems

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State of the art

- ▶ Probabilistic
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Exhaustive analysis
(precise & long)

Formula
(imprecise & fast)

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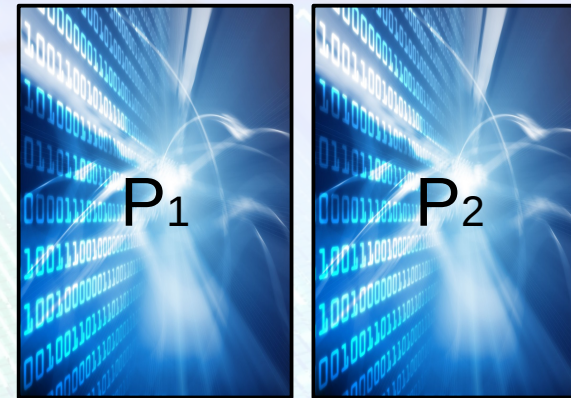
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- Probabilistic
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Distribution of
probabilities

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Static probabilistic reference method [Alt2014]

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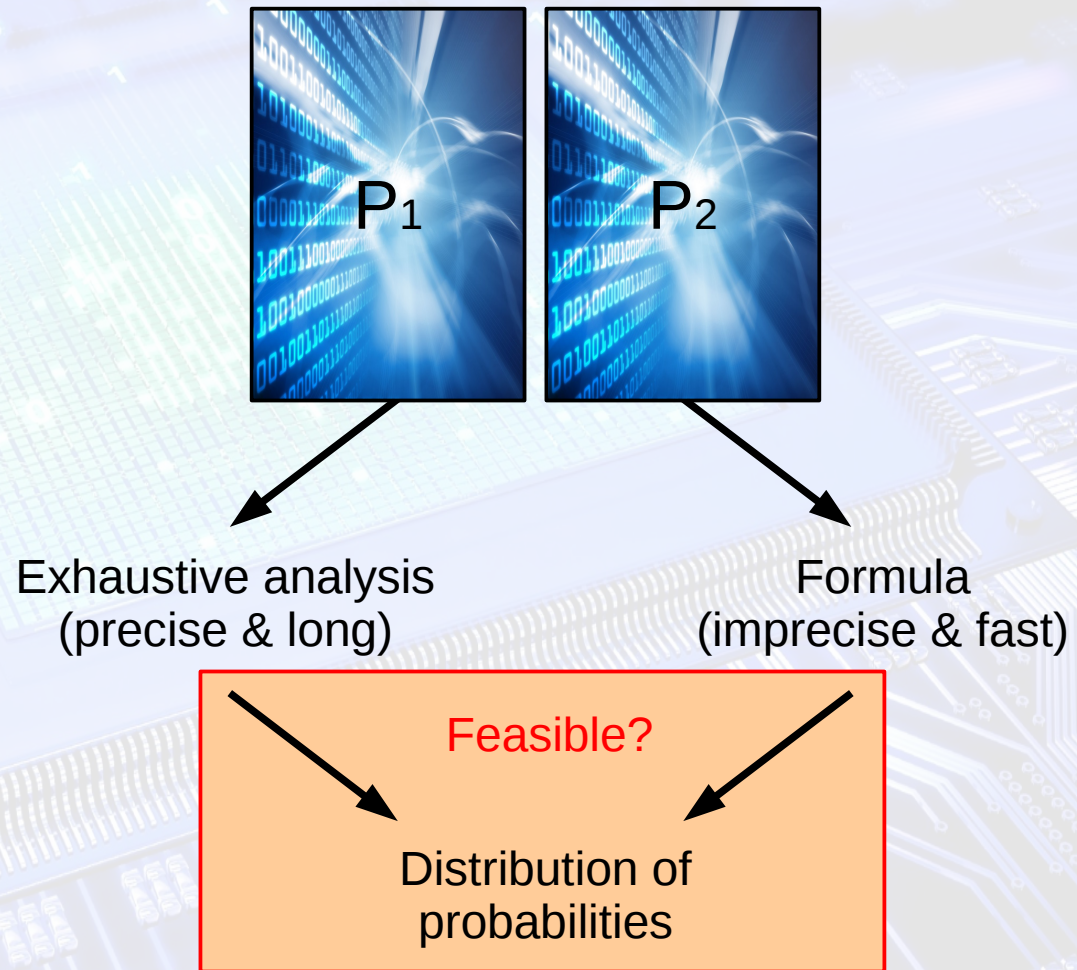
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Real-time systems

Estimation techniques

State of the art

- Probabilistic
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Real-time systems

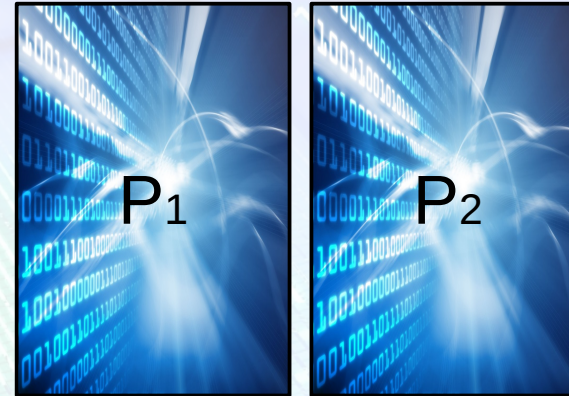
Estimation techniques

State of the art

- ▶ Probabilistic
- ▶ Static
- ▶ Low-level
- ▶ Single-path

Objectives:

- ▶ Same context
- ▶ Safe & tighter



Exhaustive analysis
(precise & long)

Formula
(imprecise & fast)

Feasible?

Distribution of
probabilities

[Alt2014] S. Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis" (2014)

Outline

- 1°) Required background
- 2°) Proposed method
- 3°) Experimental evaluation
- 4°) Conclusion & Future work

Outline

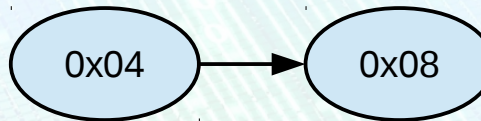
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Fully-associative cache memories

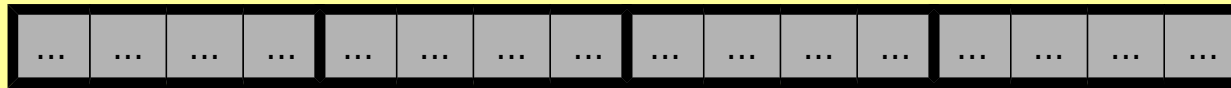
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Cache memories
Distributions
Convolution

- ▶ Small & fast memories
- ▶ Found in most architectures
- ▶ Takes advantage of the locality



Cache
(~2 cycles)



Replacement
policy

Main memory
(~70 cycles)

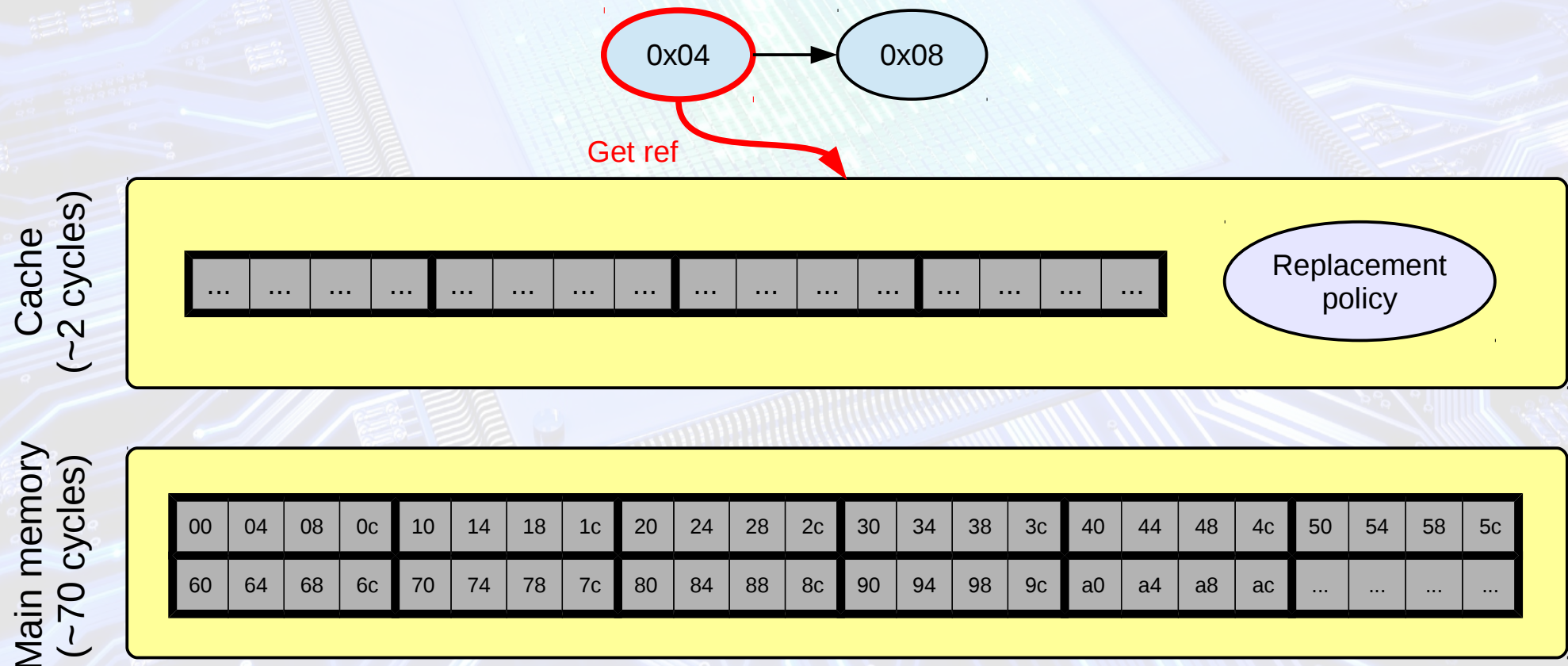
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60	64	68	6c	70	74	78	7c	80	84	88	8c	90	94	98	9c	a0	a4	a8	ac	...	...	...	...

Fully-associative cache memories

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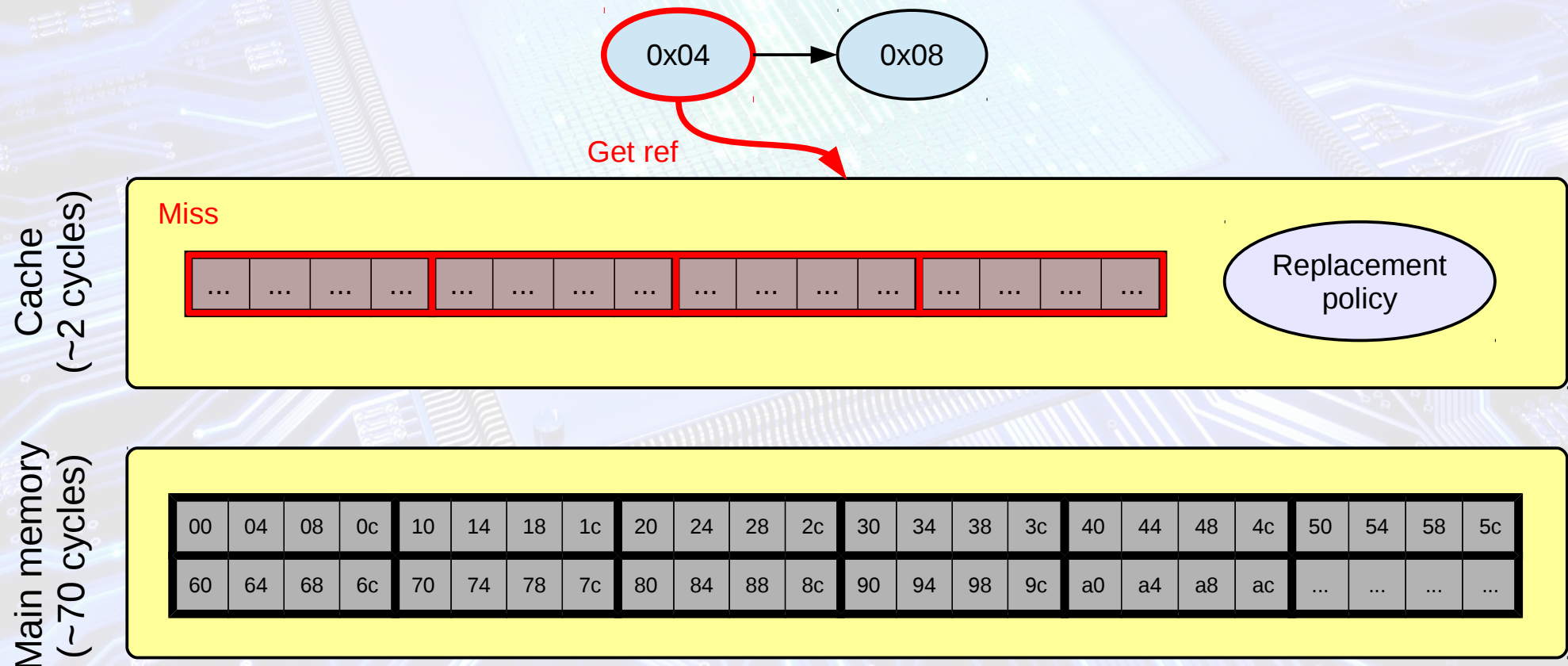


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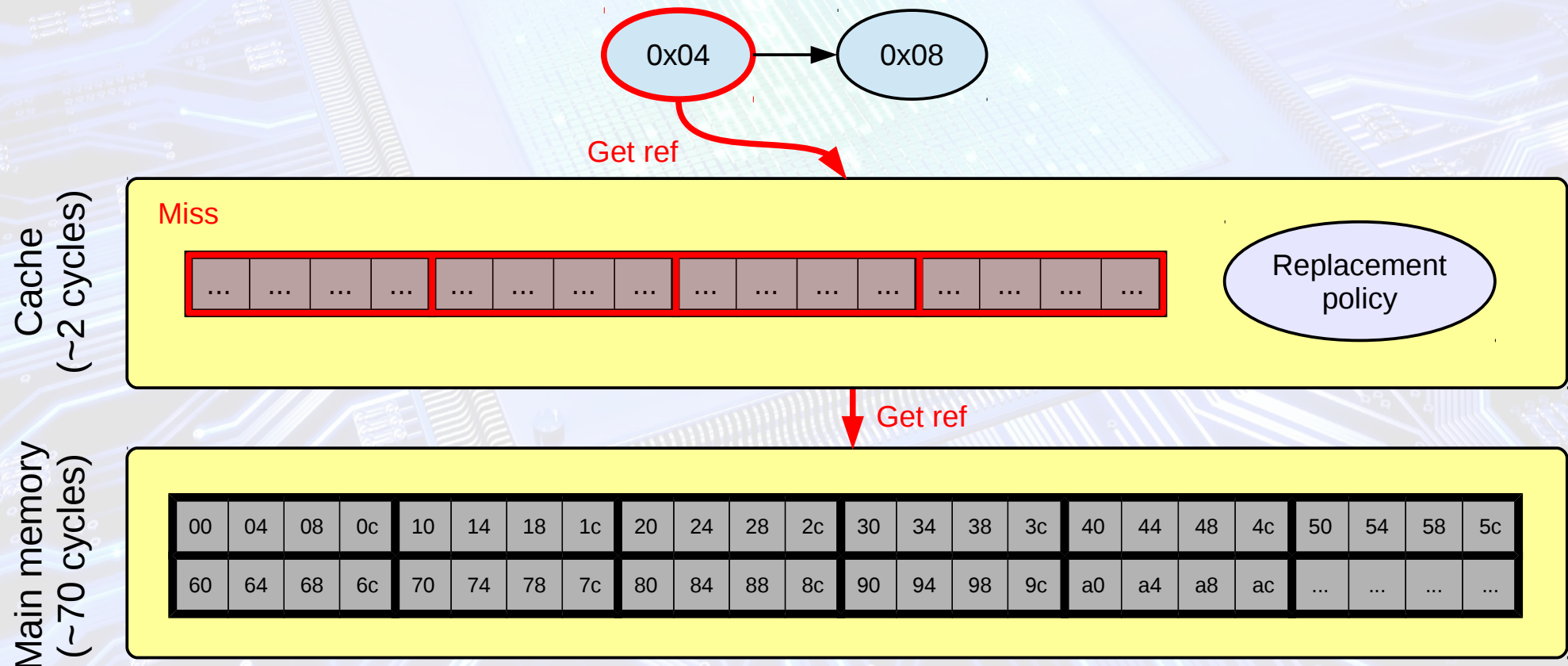


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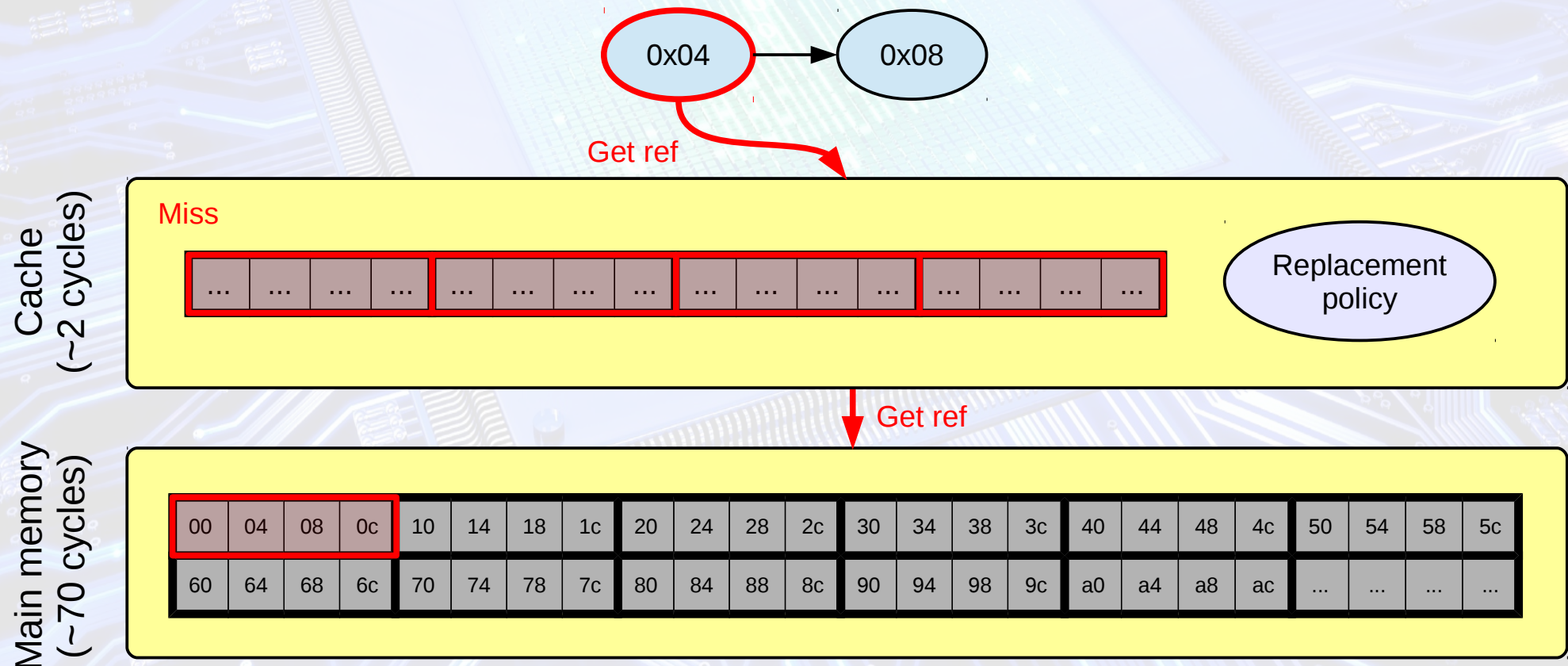


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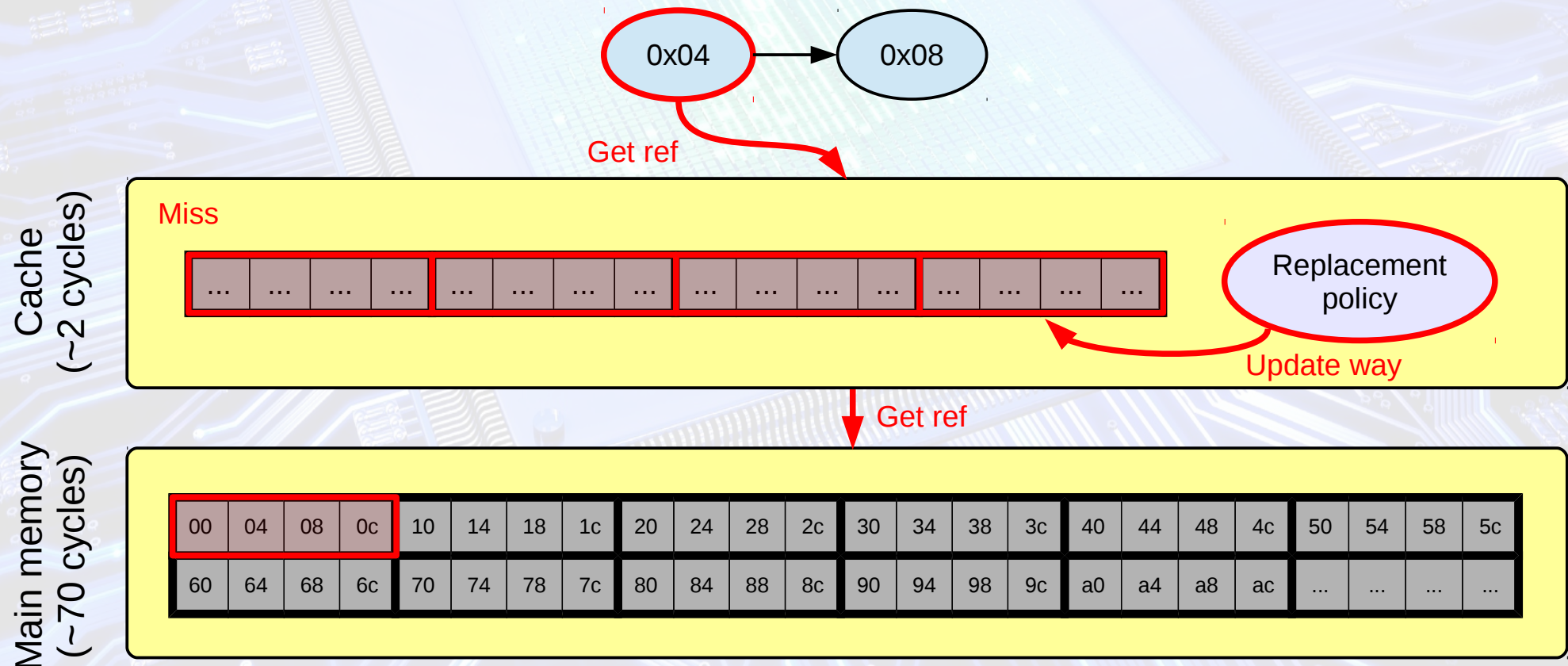


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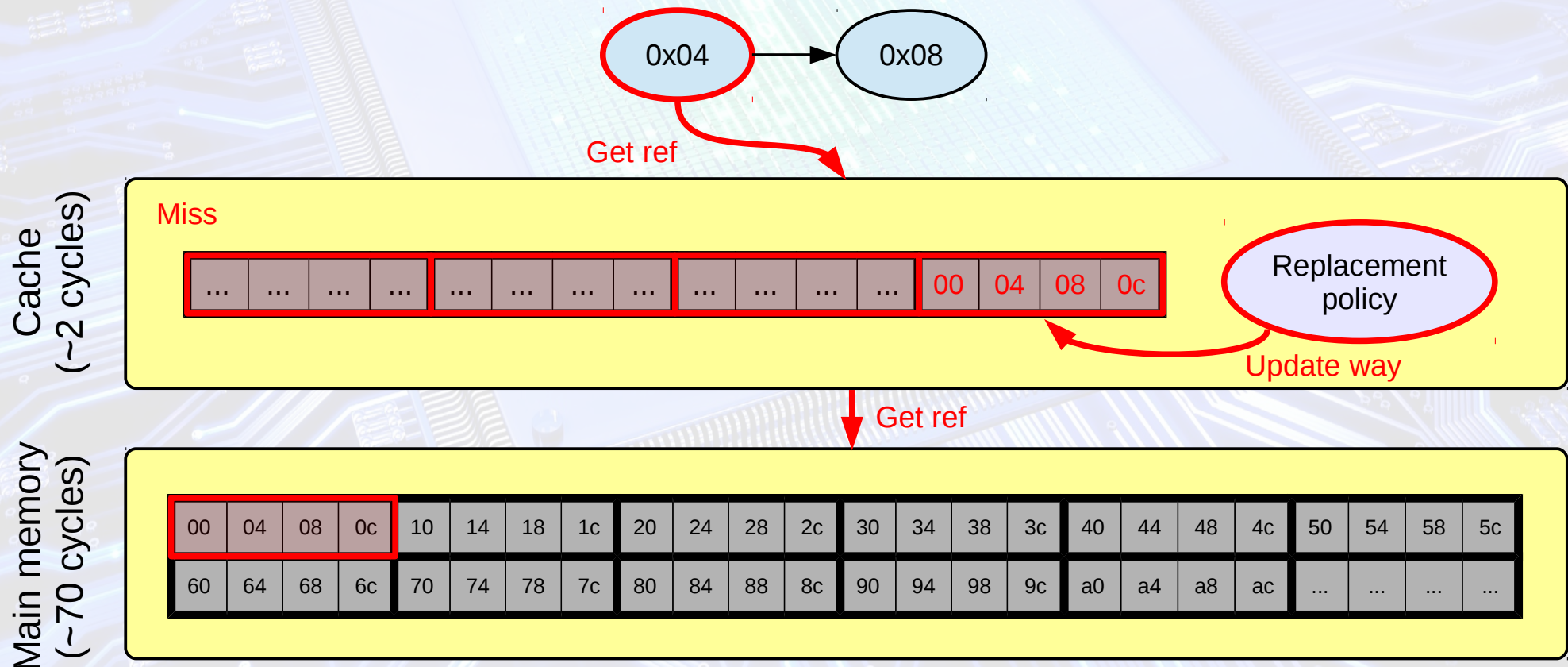


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Fully-associative cache memories

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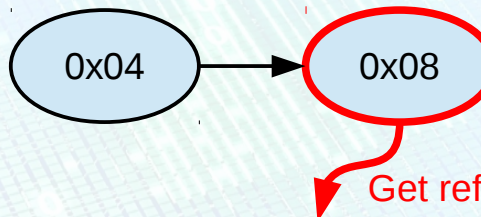
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Cache memories

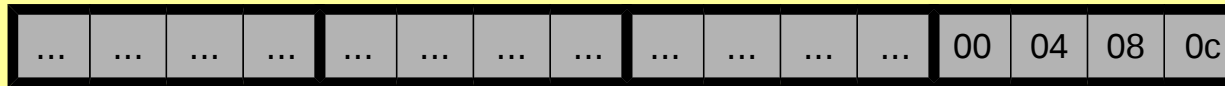
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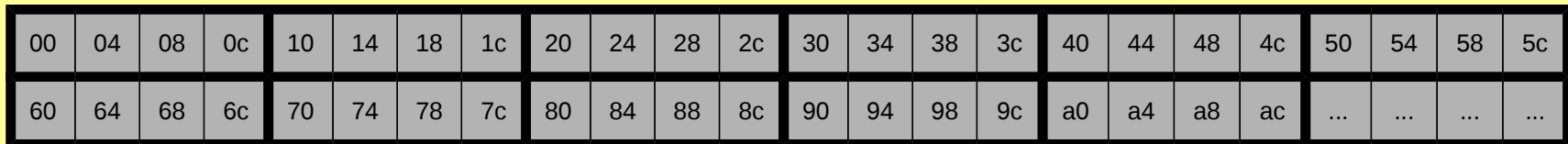


Cache
(~2 cycles)



Replacement
policy

Main memory
(~70 cycles)

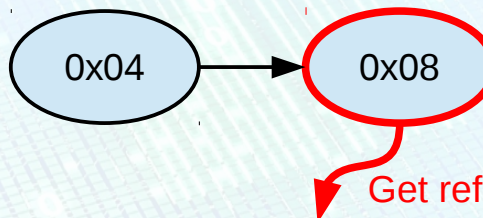


Fully-associative cache memories

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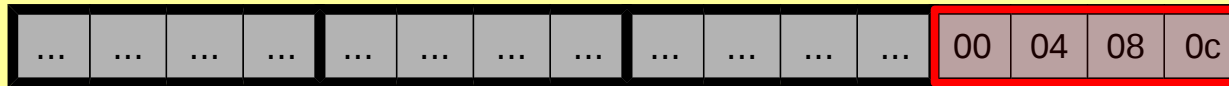
Cache memories
Distributions
Convolution

- ▶ Small & fast memories
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- ▶ Takes advantage of the locality



Cache
(~2 cycles)

Hit



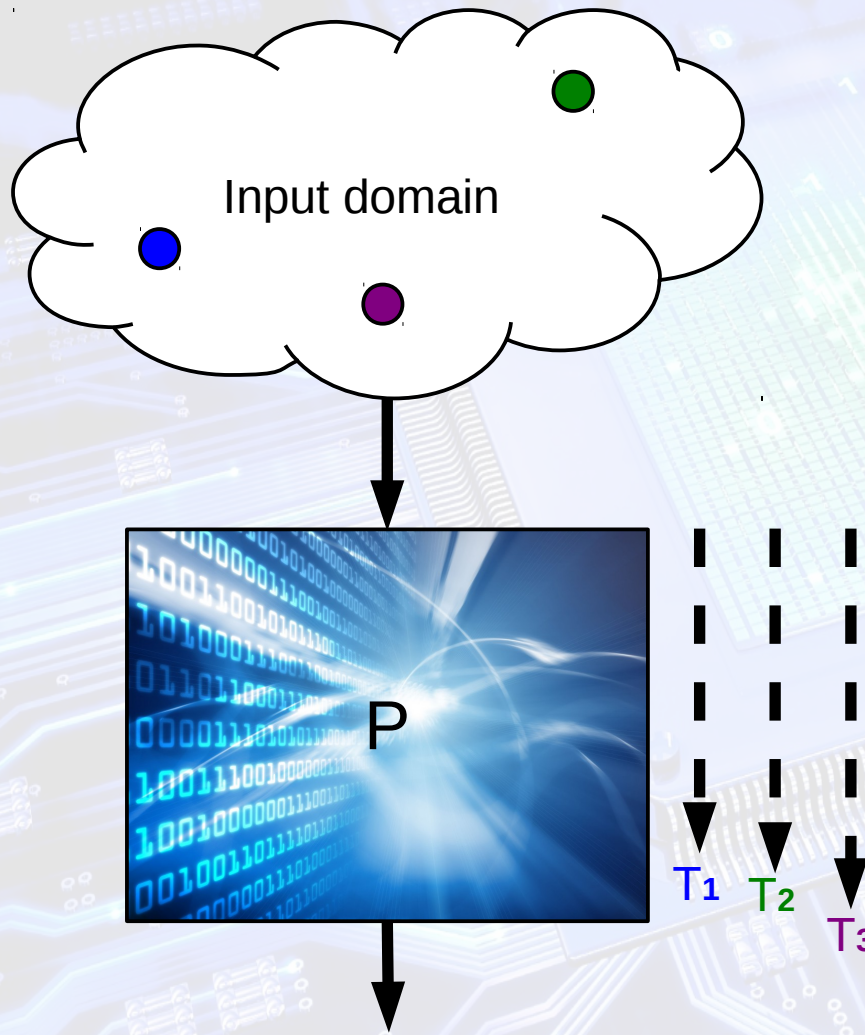
Main memory
(~70 cycles)

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Distributions of timing values

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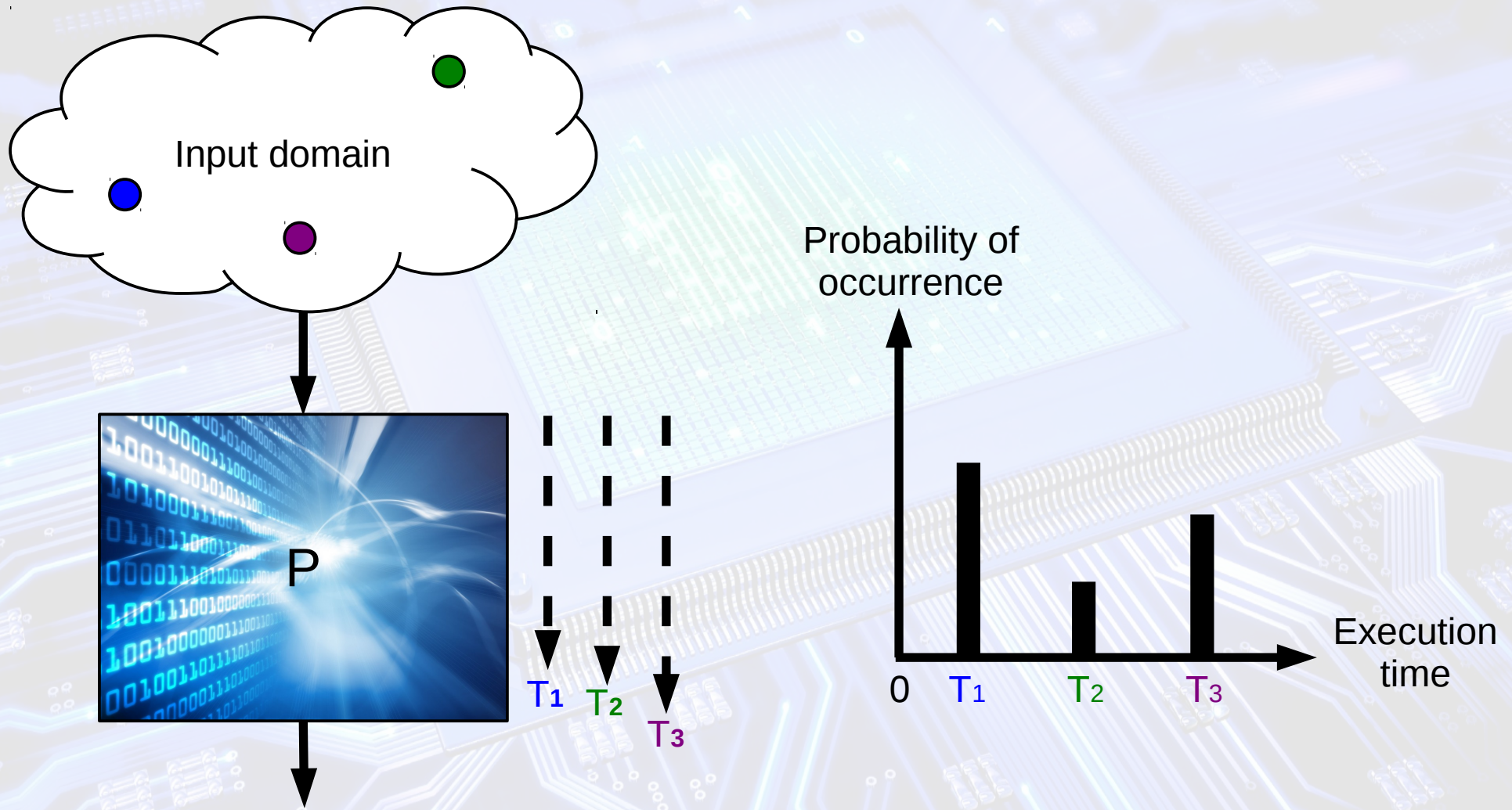
Cache memories
Distributions
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Distributions of timing values

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Cache memories
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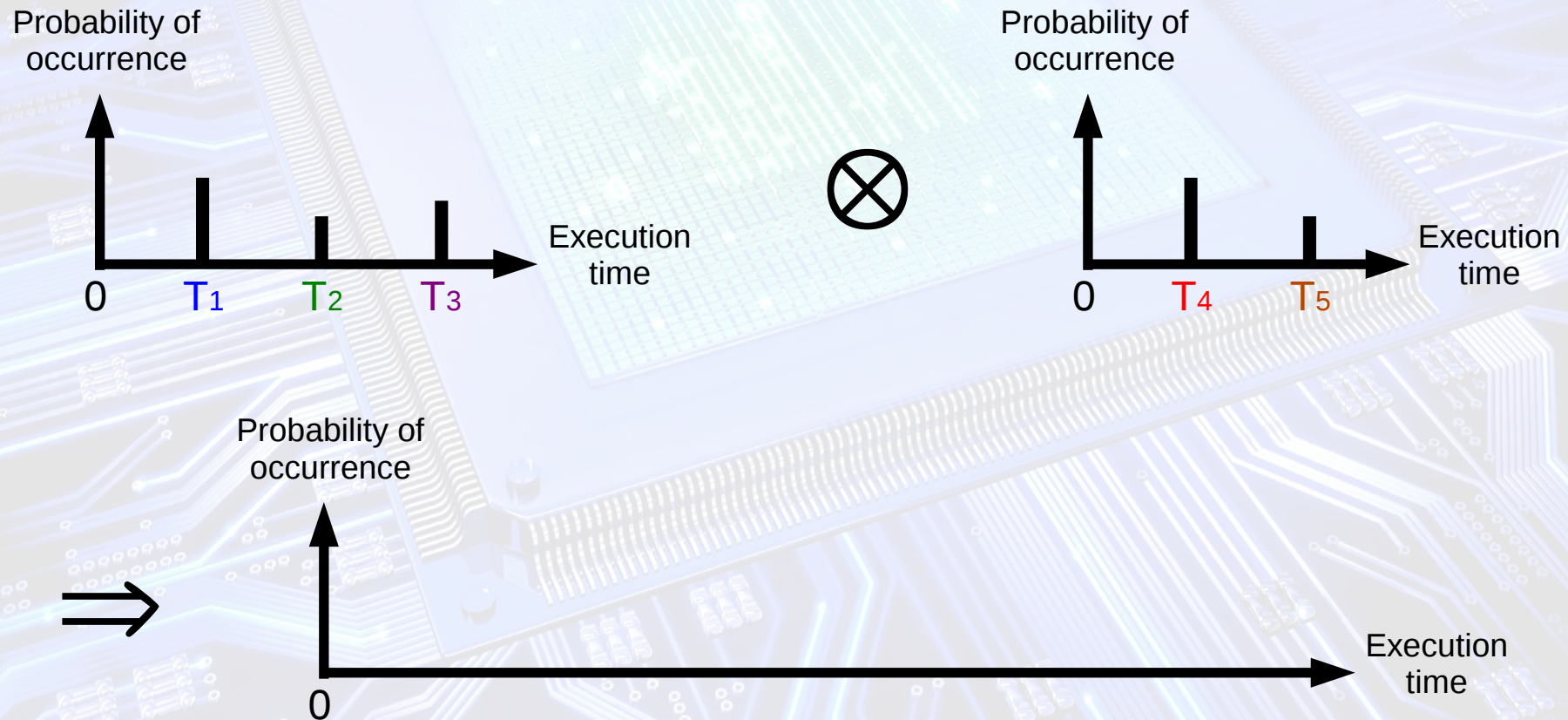


Product of convolution

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Cache memories
Distributions
Convolution

- Requires **independence** between distributions

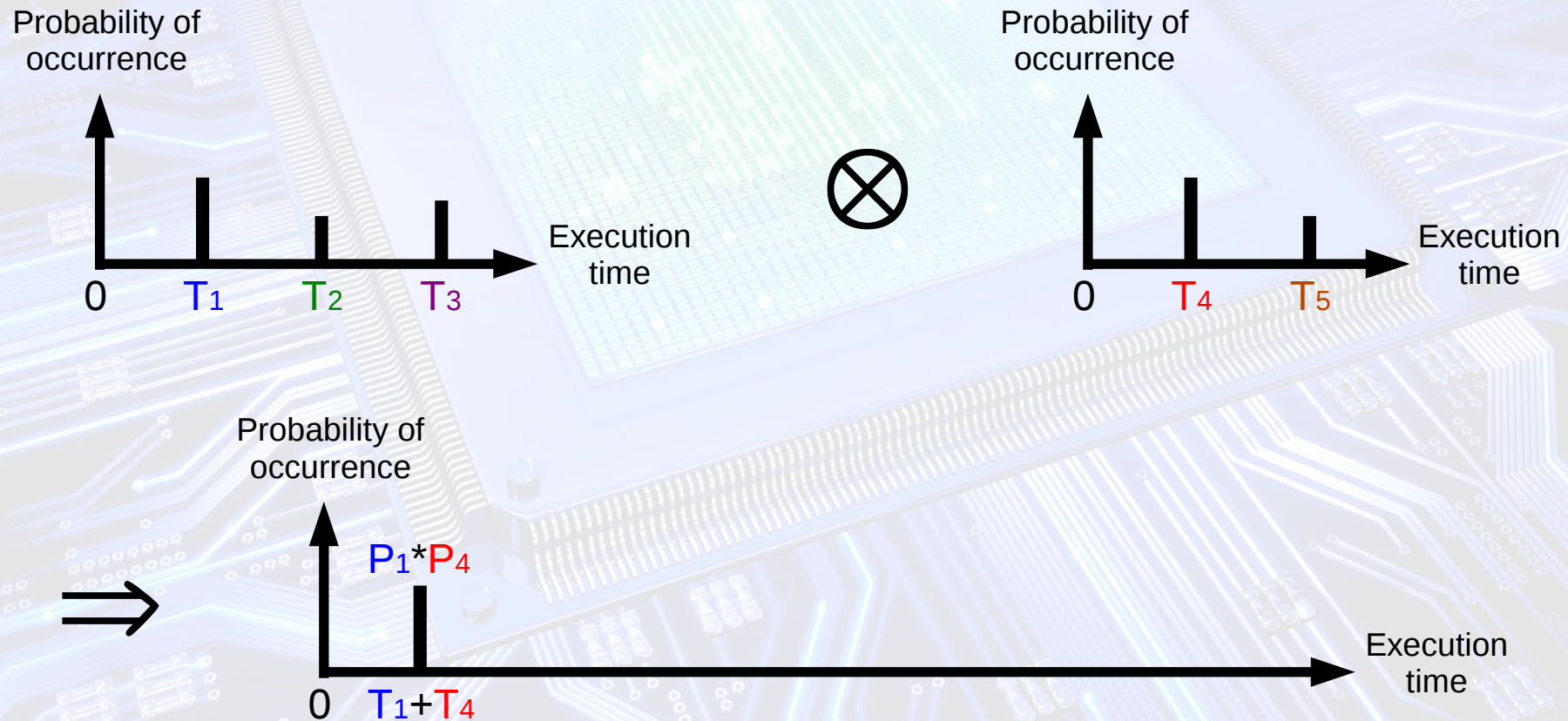


Product of convolution

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Cache memories
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- Requires **independence** between distributions

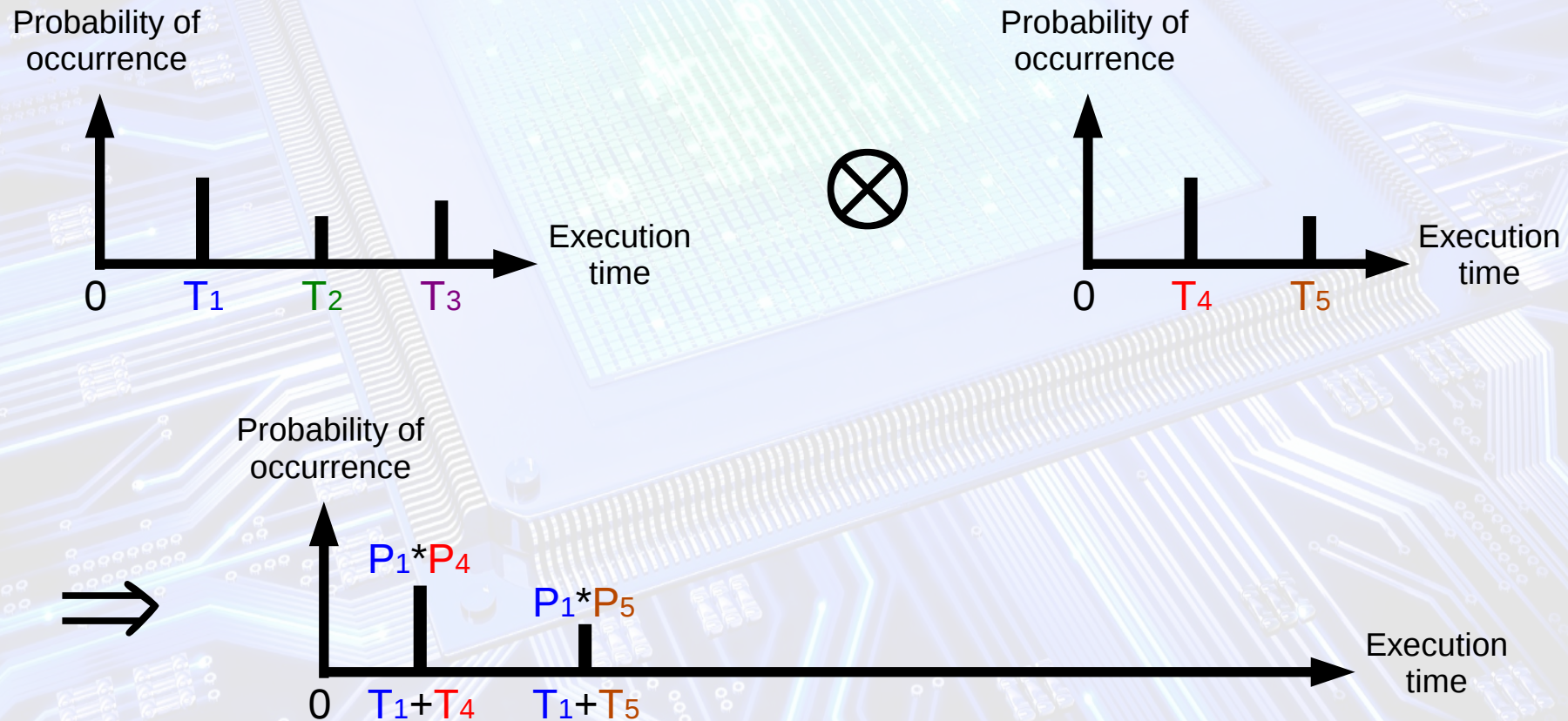


Product of convolution

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- Requires **independence** between distributions

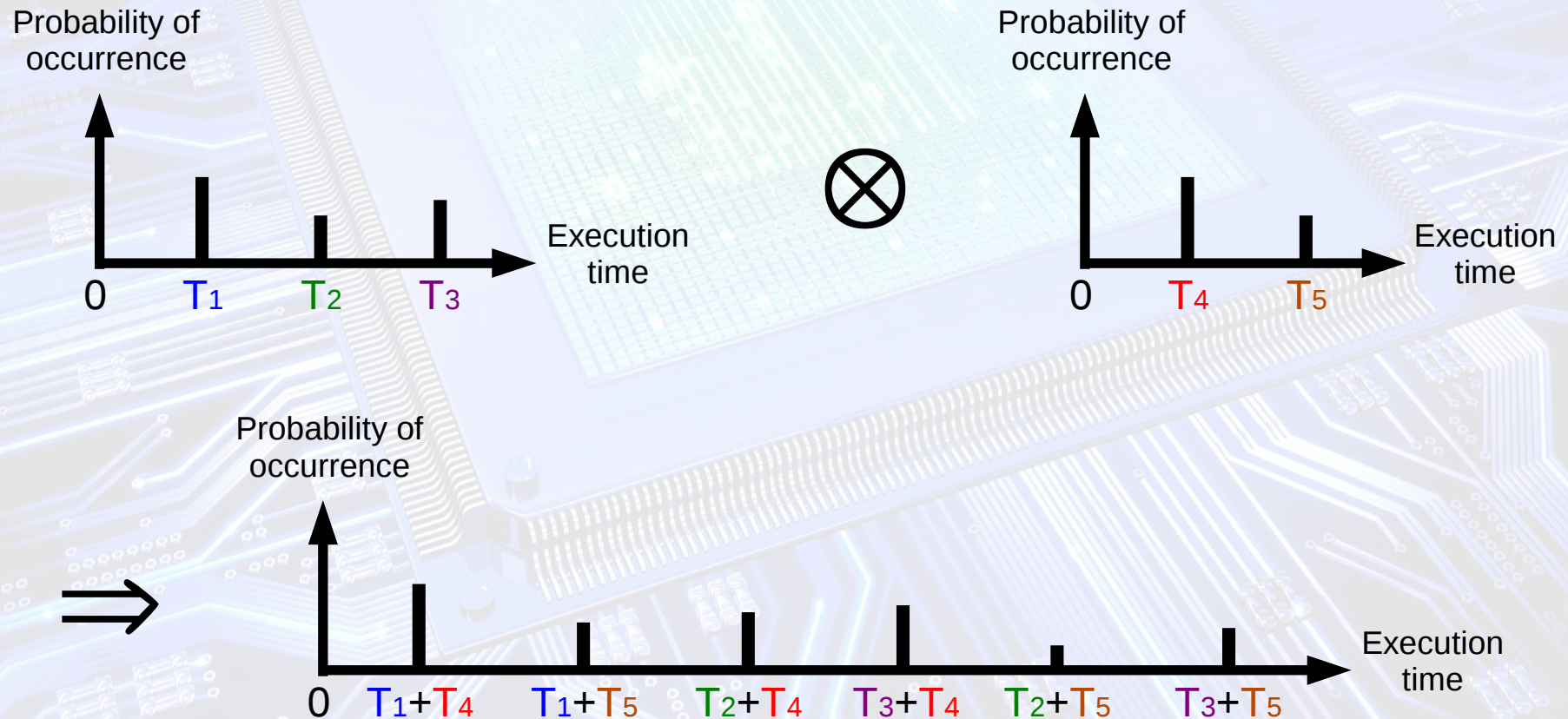


Product of convolution

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Cache memories
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- Requires **independence** between distributions



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- 1°) Required background
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Hardware & software

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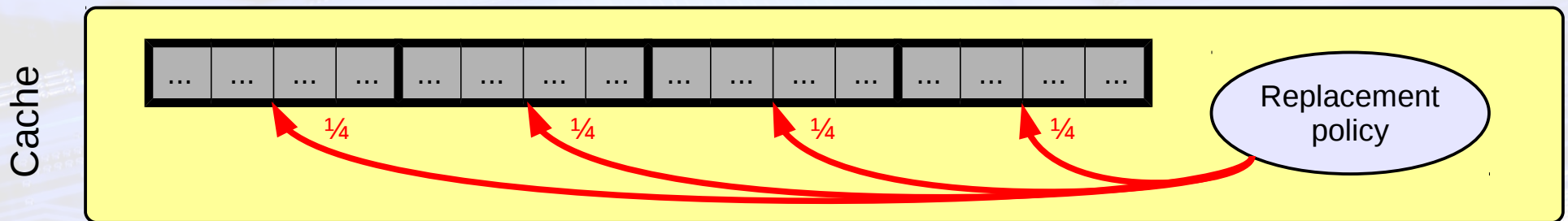
Context

Exhaustive enumeration

Reduction of the problem

Experimental evaluation

► Random replacement caches (4 ways)

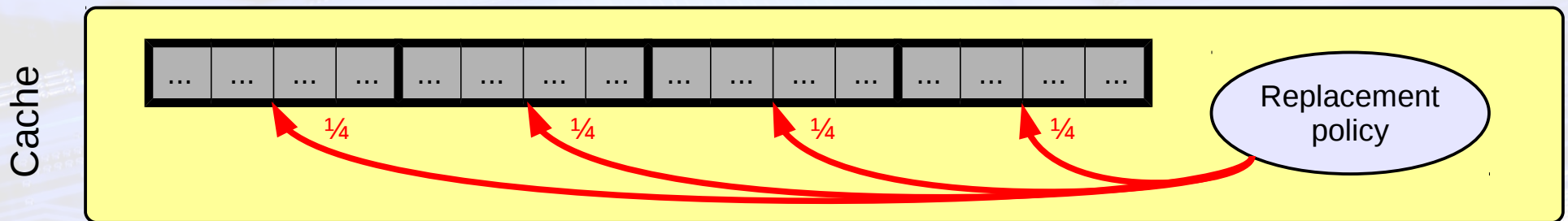


Hardware & software

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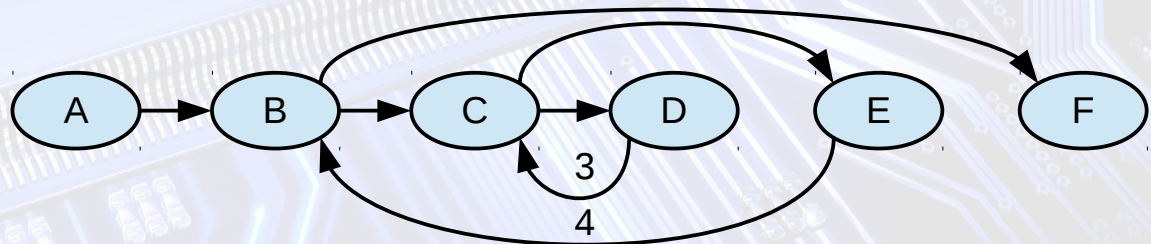
Context
Exhaustive enumeration
Reduction of the problem
Experimental evaluation

► Random replacement caches (4 ways)



► Single-path programs

```
FOR I = 1 TO 4 :  
  FOR J = 1 TO 3 :  
    ...
```



ABCD CDCDCEBCDCDCDCDCEBCDCDCDCDCEBCDCDCDCDCEBF

Exhaustive cache states enumeration

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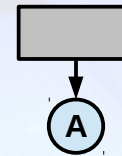
ABCD...C...

Exhaustive cache states enumeration

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Experimental evaluation

(A)BCDC...



Exhaustive cache states enumeration

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Exhaustive enumeration

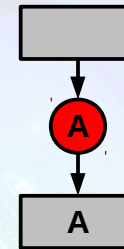
Proposed method

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(A)BCDC...



Exhaustive cache states enumeration

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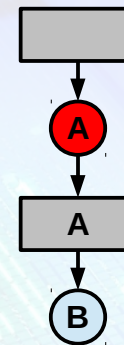
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A(B)CDC...

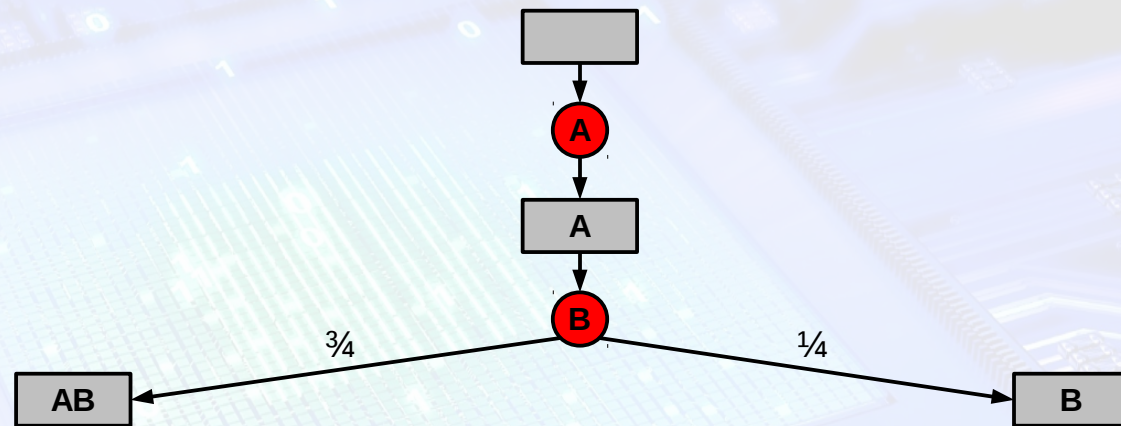


Exhaustive cache states enumeration

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A(B)CDC...

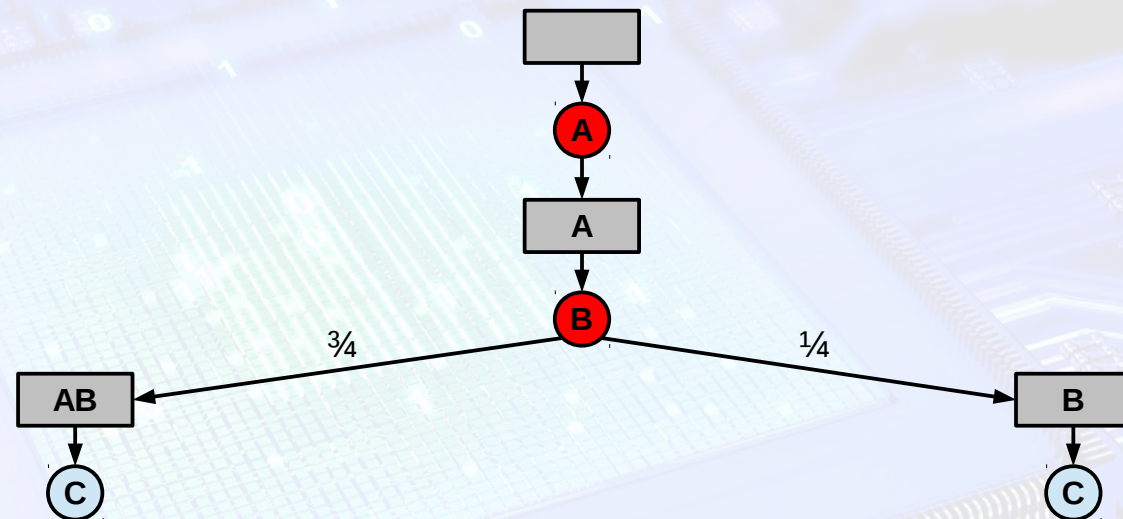


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AB(C)DC...

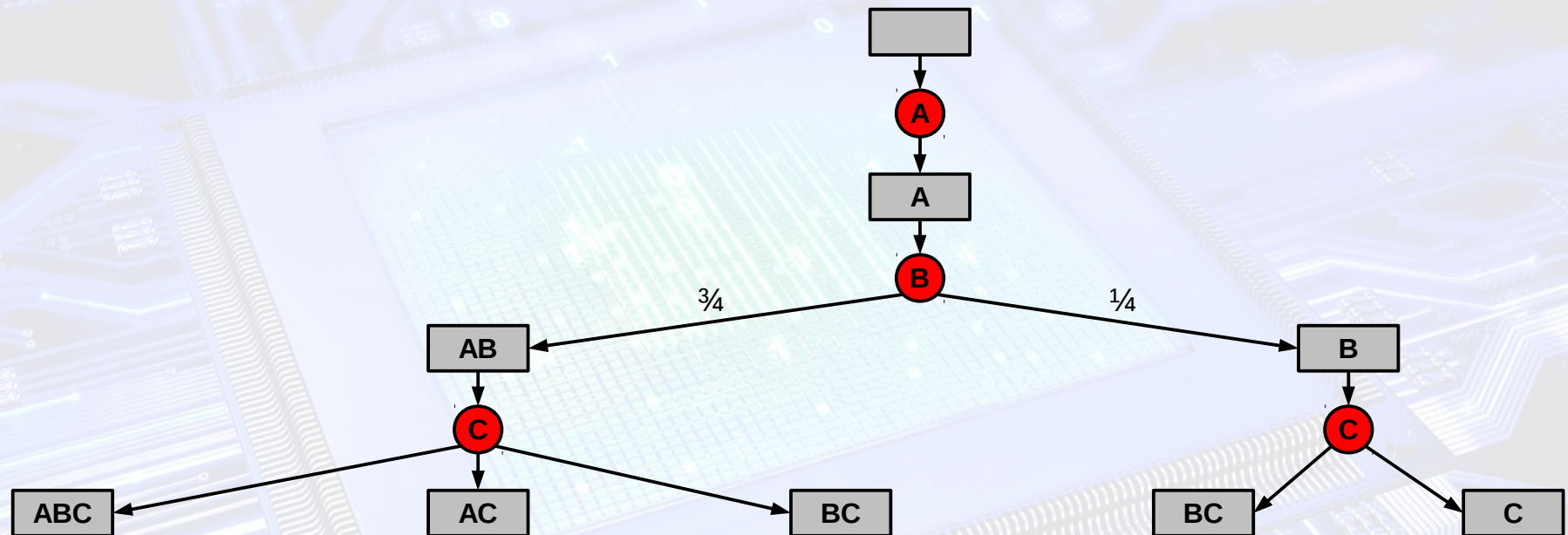


Exhaustive cache states enumeration

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Experimental evaluation

AB(C)DC...

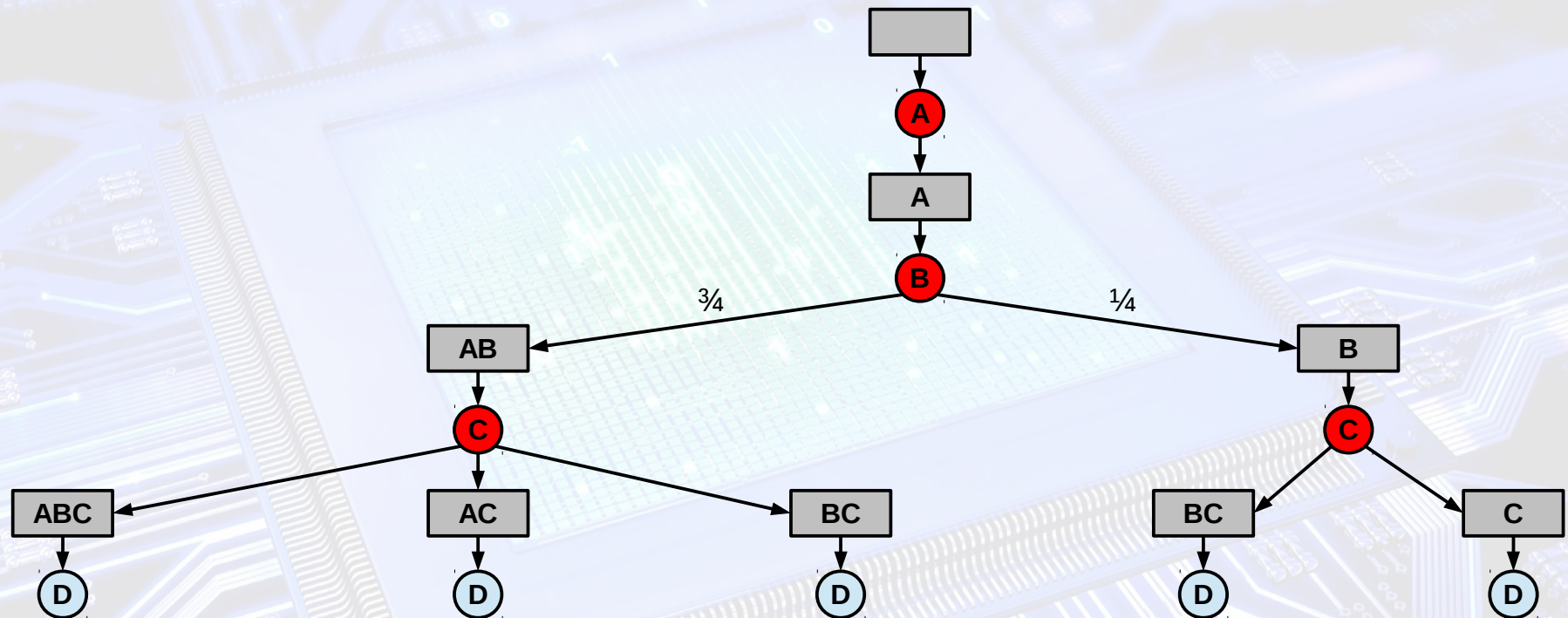


Exhaustive cache states enumeration

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ABC(D)C...

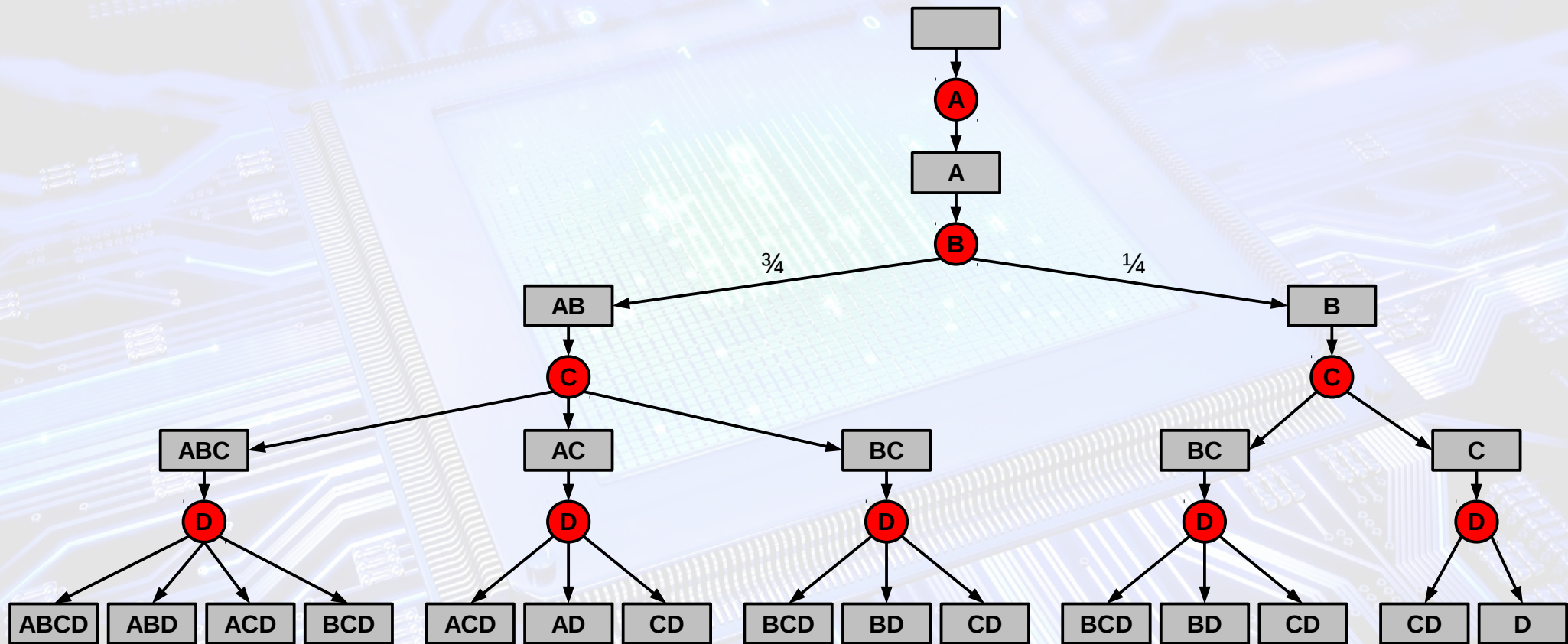


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ABC(D)C...



Exhaustive cache states enumeration

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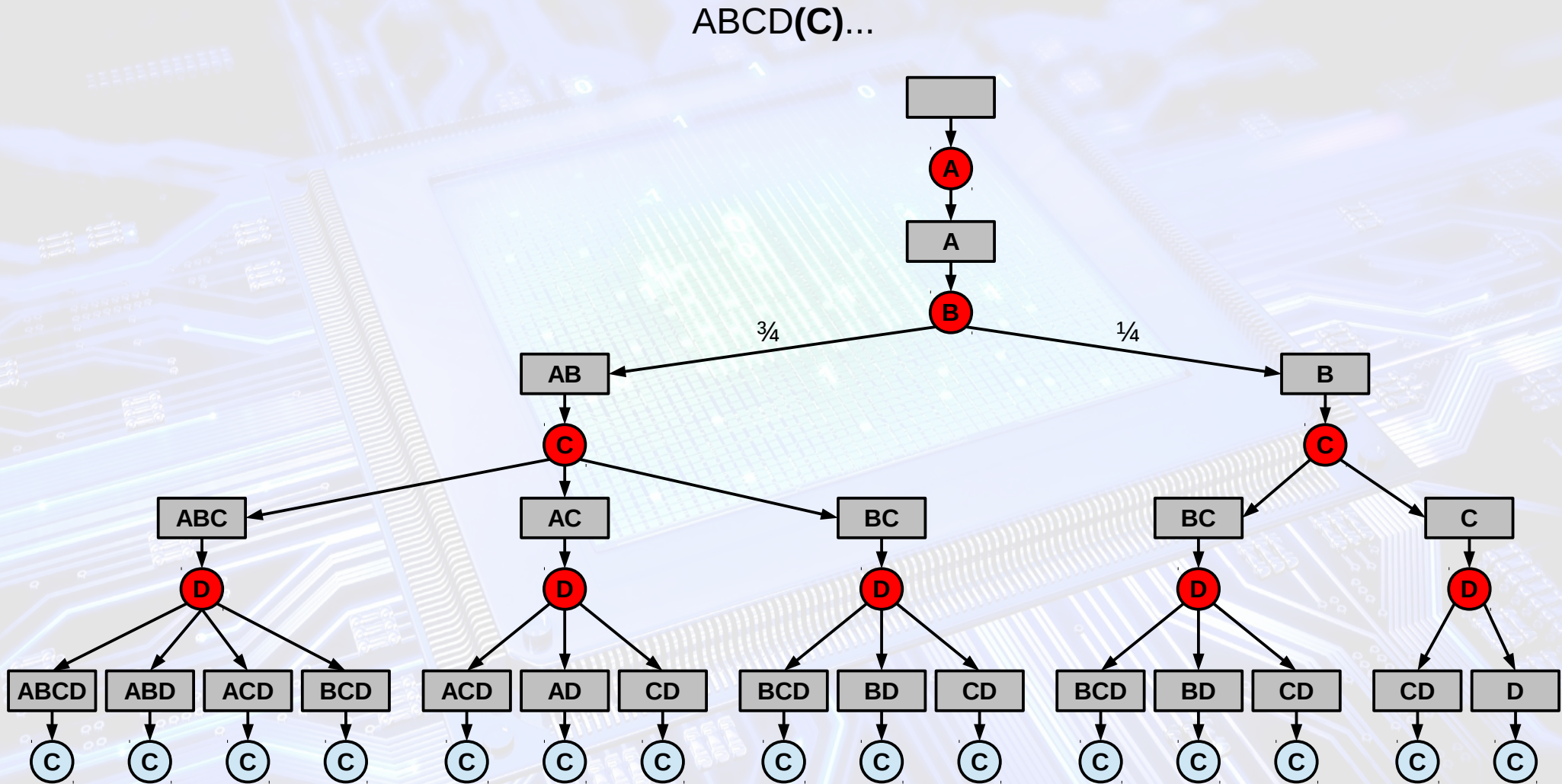
Exhaustive enumeration

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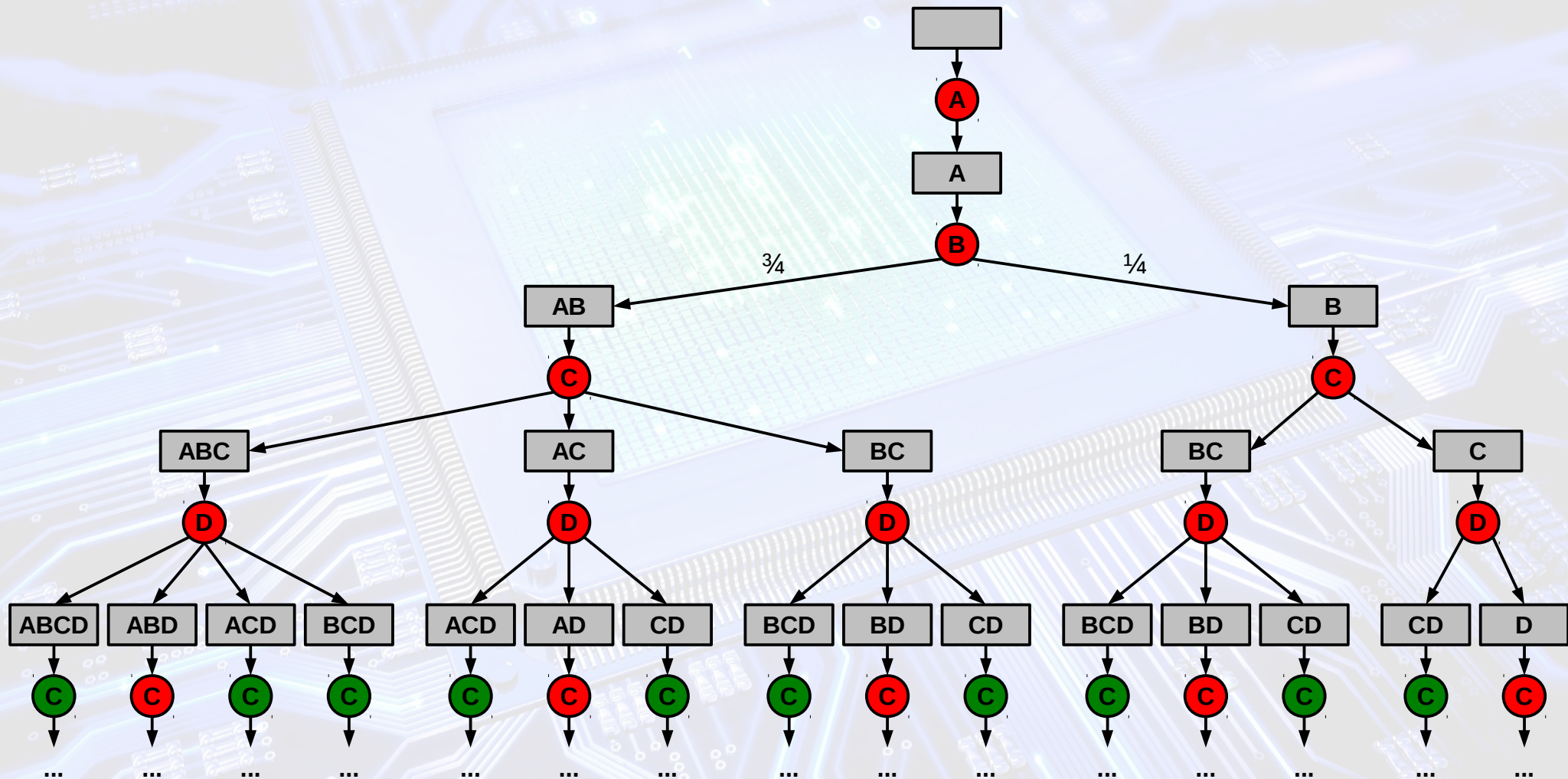


Exhaustive cache states enumeration

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ABCD(C)...

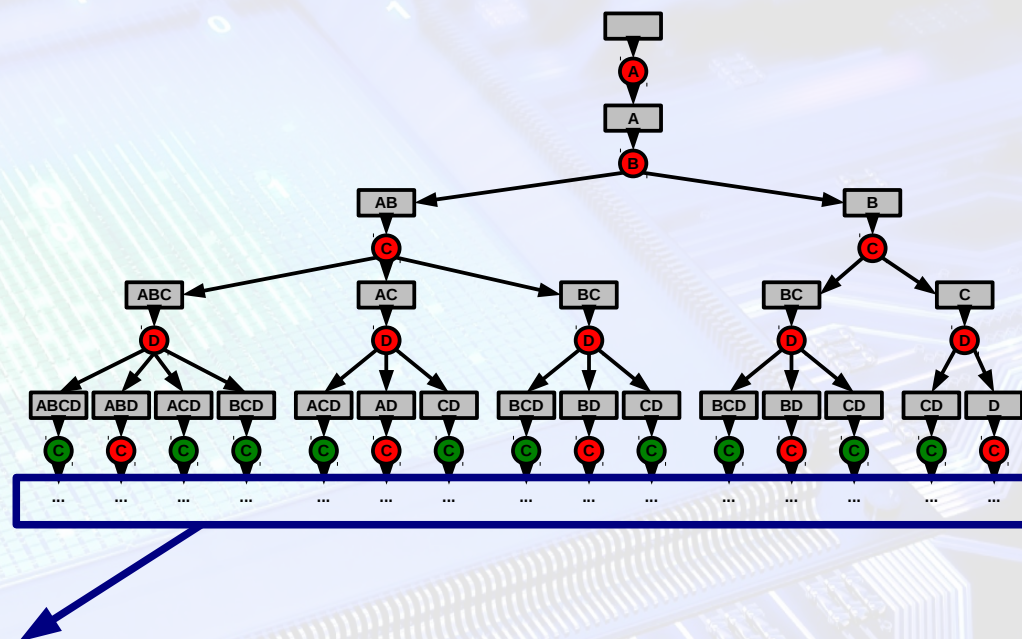
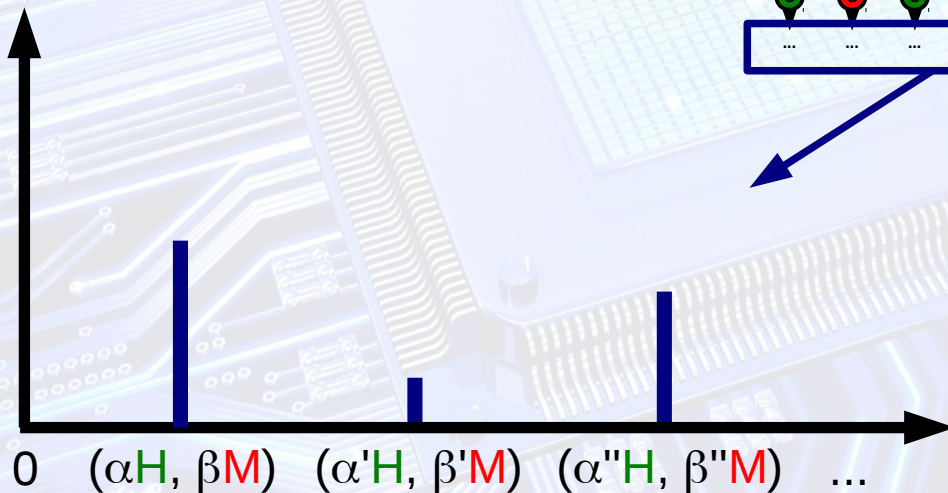


Computation of a distribution from the leaves

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Probability of occurrence



Execution time

Pros & cons

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- ▶ Exponential complexity
- ▶ Intractable



- ▶ Real probabilities
- ▶ Exact distribution

Pros & cons

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- ▶ Exponential complexity
- ▶ Intractable



- ▶ Real probabilities
- ▶ Exact distribution

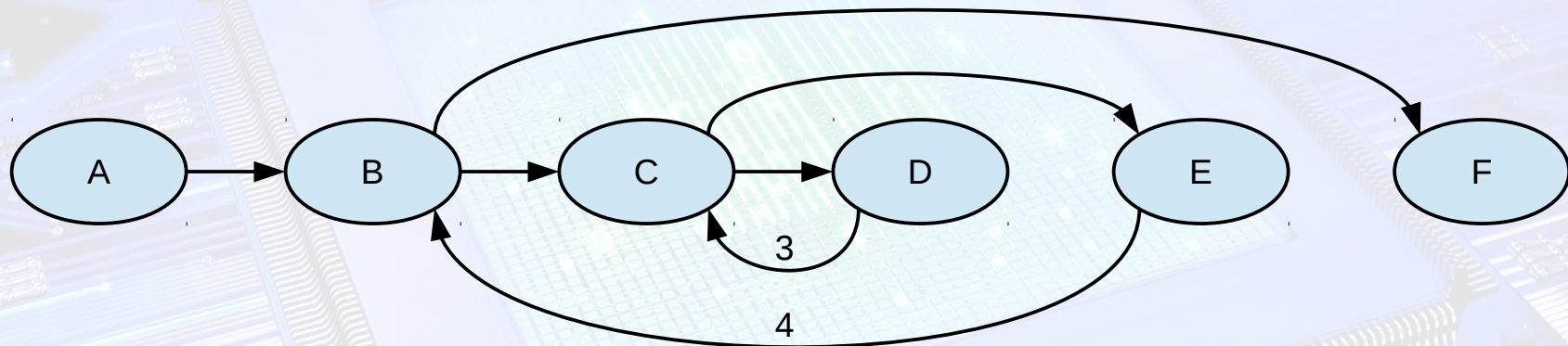
Idea:

- ▶ Cut into windows
- ▶ Exhaustive analyses
- ▶ Convolution

Program under study

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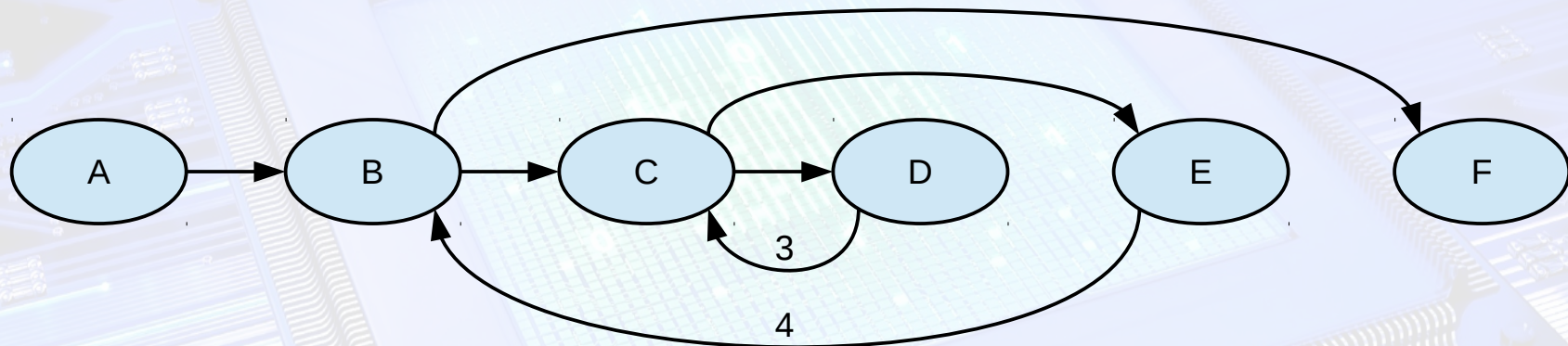


ABCD CDCDCEBCDCDCDCEBCDCDCDCEBCDCDCDCEBF

A naive and inefficient cut

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[ABCD CDCDCEBCDCDCDCE][BCDCDCDCEBCDCDCDCEBF]

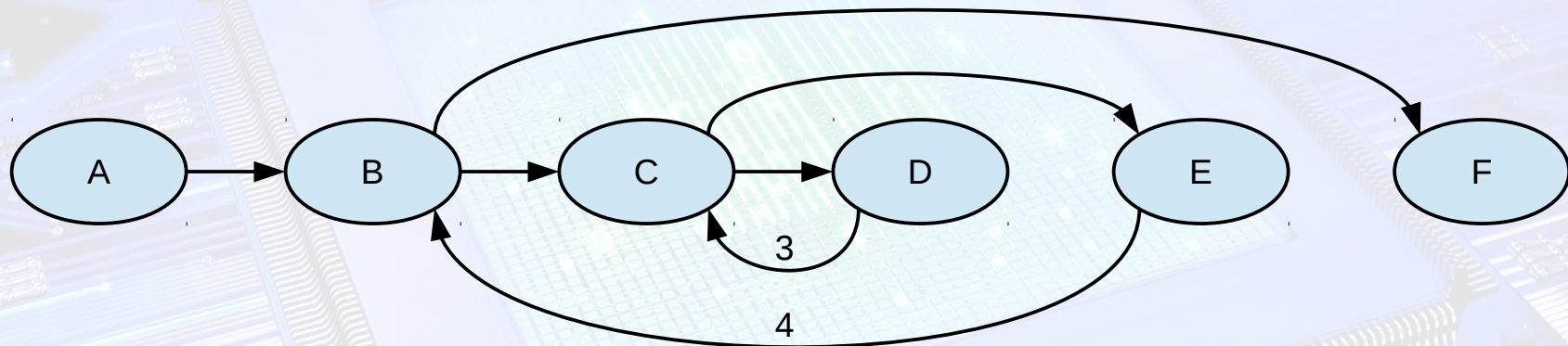
W potential misses W potential misses

Cut in two

A naive and inefficient cut

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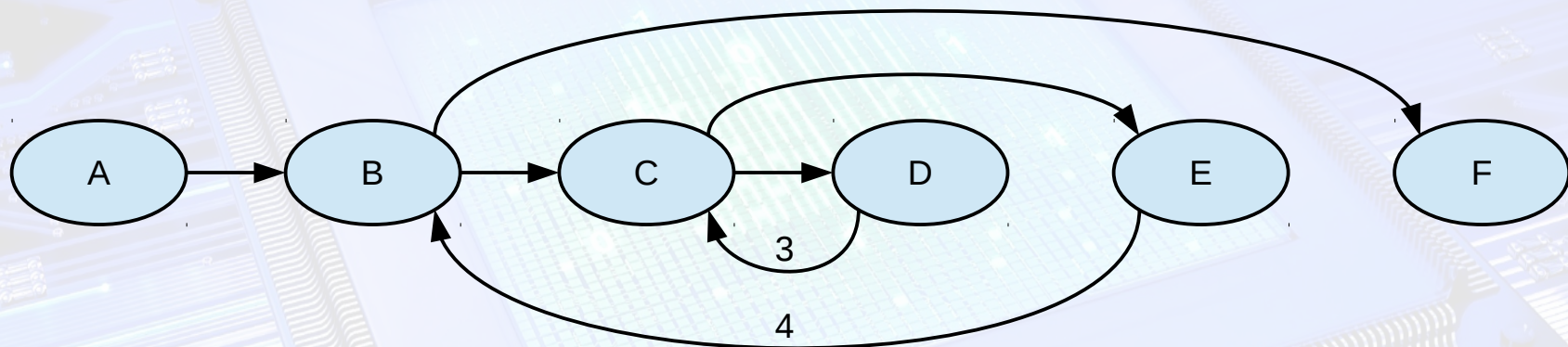
ABCD CDCDCEBCDCDCDCE

BCDCDCDCEBCDCDCDCEBF

A naive and inefficient cut

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ABCD CDCDCEBCDCDCDCE

Safe
cache

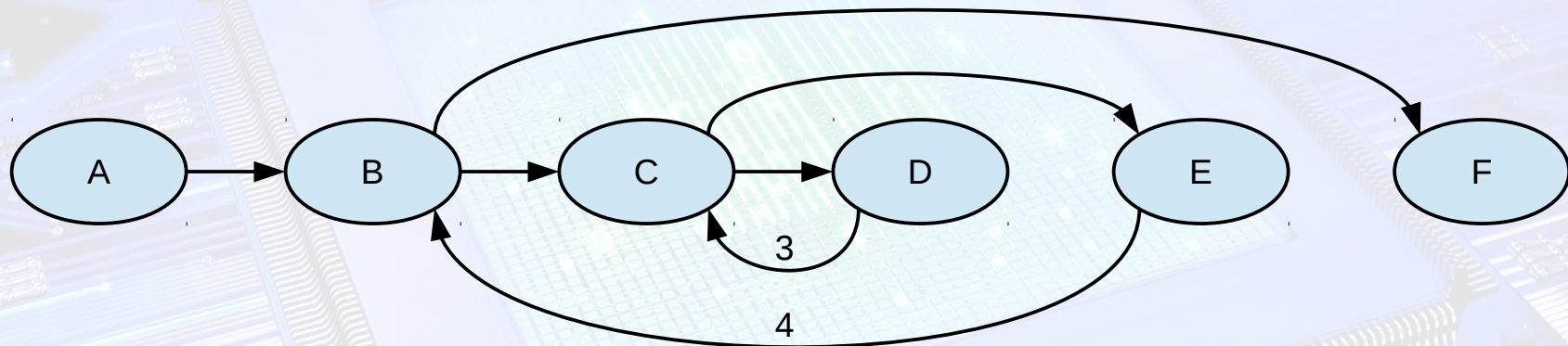
BCDCDCDCEBCDCDCDCEBF

Safe
cache

Reuse comes from the loops

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Experimental evaluation

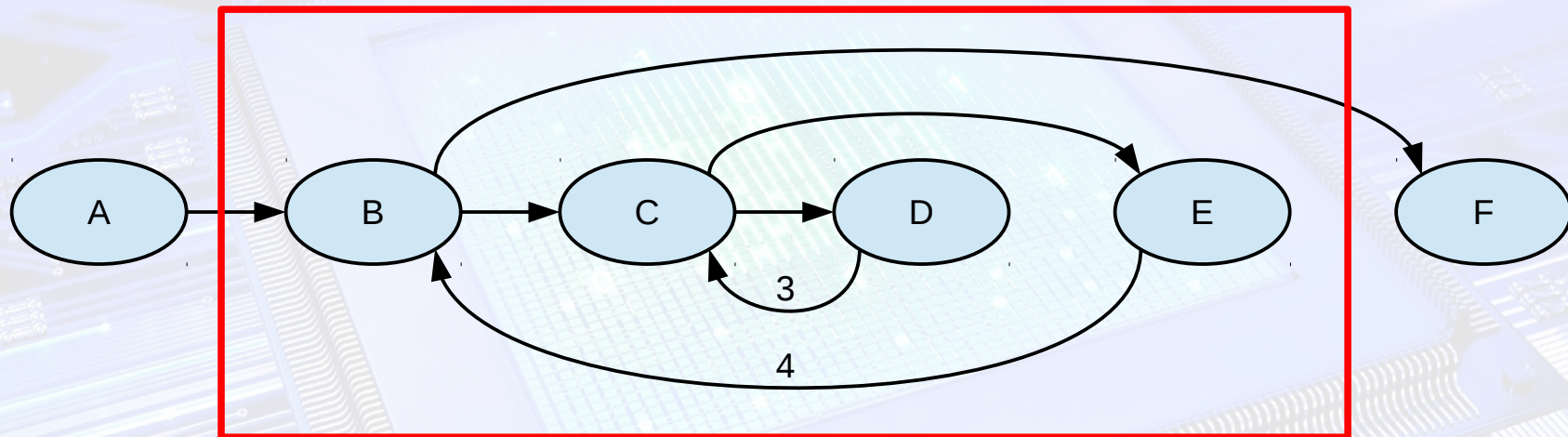


ABCD CDCDCEBCDCDCDCEBCDCDCDCEBCDCDCDCEBF

Reuse comes from the loops

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A

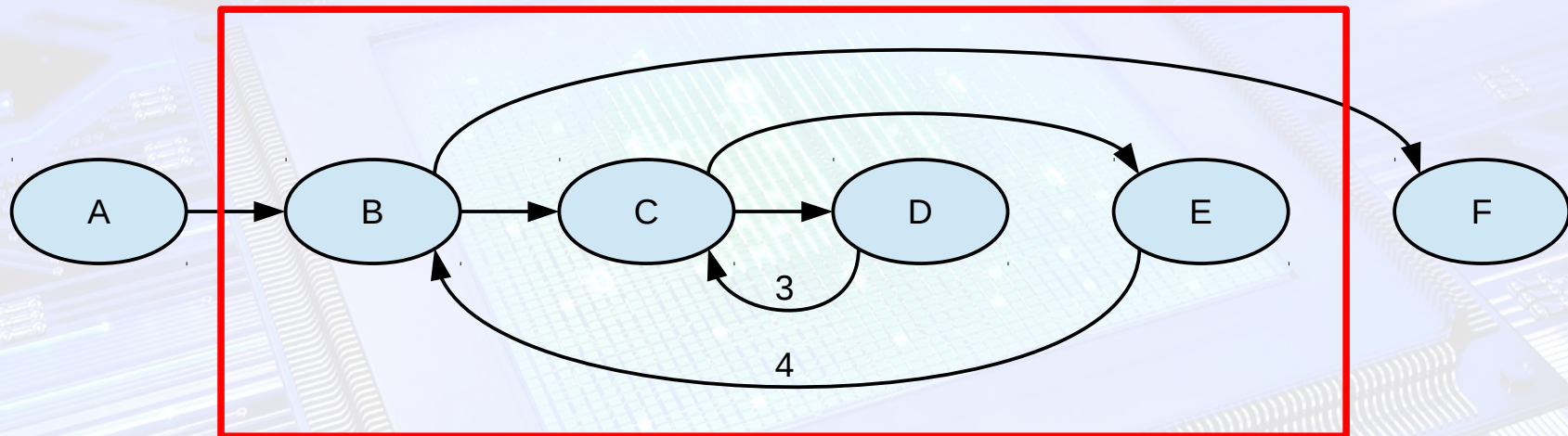
[BCDCDCDCEBCDCDCDCEBCDCDCDCEBCDCDCDCEB]

F

Reuse comes from the loops

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Reduction of the problem
Experimental evaluation



A

[BCDCDCDCEBCDCDCDCEBCDCDCDCEBCDCDCDCEB]

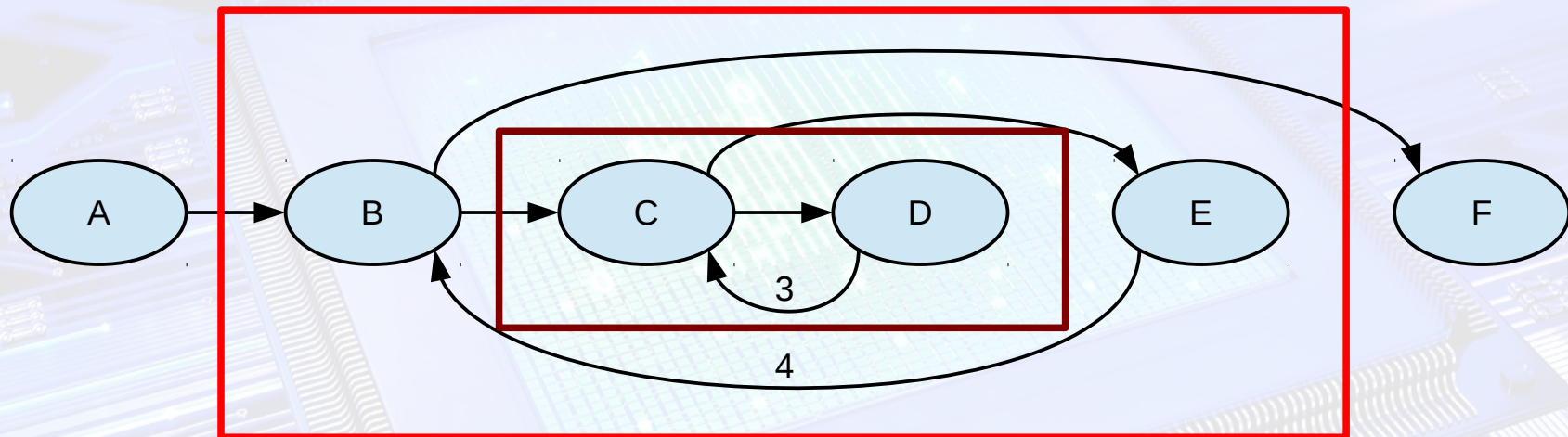
F

Still too long
($>W$ potential misses)

Reuse comes from the loops

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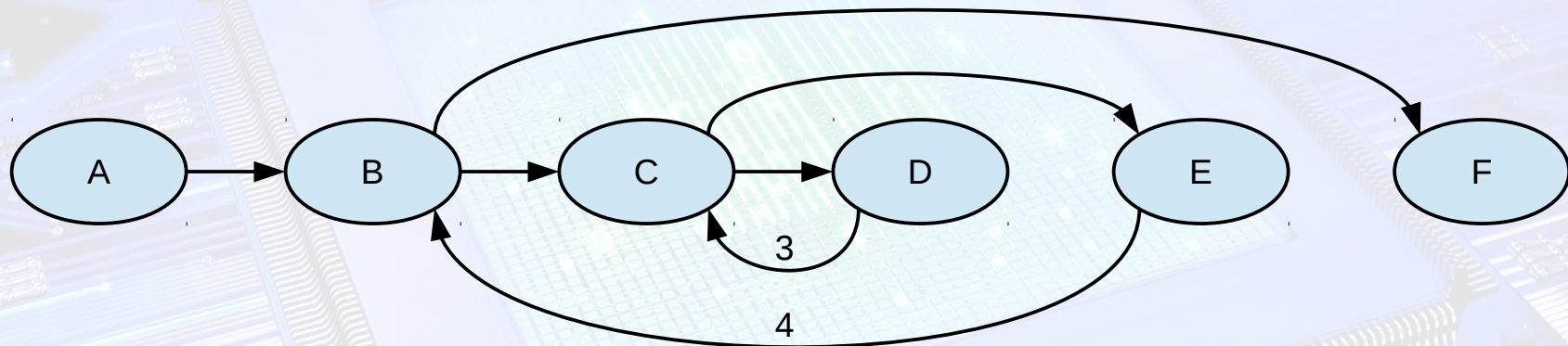


A **[[B][CDCDCDC][EB][CDCDCDC][EB][CDCDCDC][EB][CDCDCDC][EB]]** F

Reuse comes from the loops

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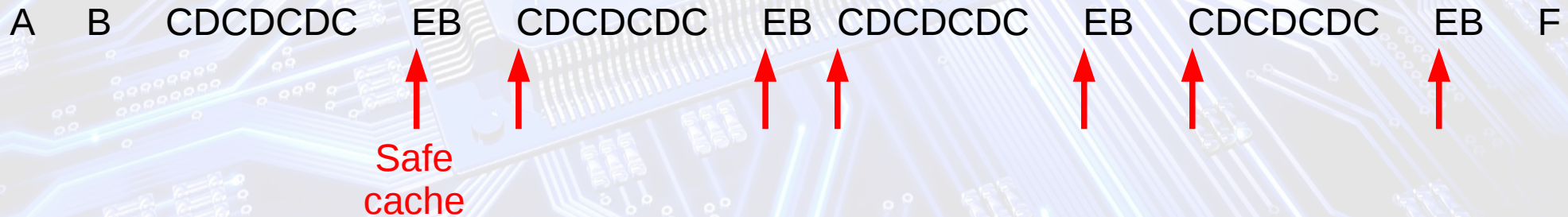
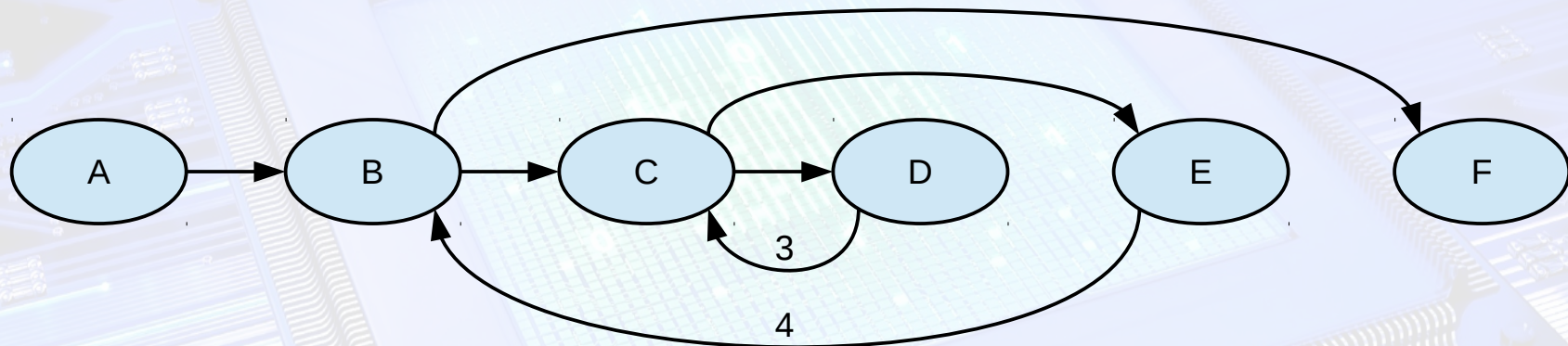


A B CDCDCDC EB CDCDCDC EB CDCDCDC EB CDCDCDC EB F

Reuse comes from the loops

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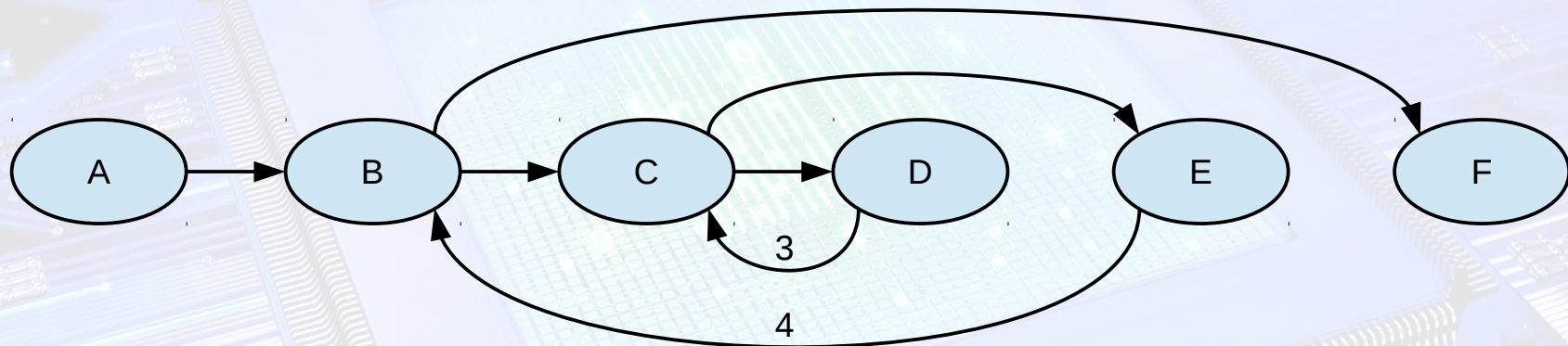
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We merge small windows

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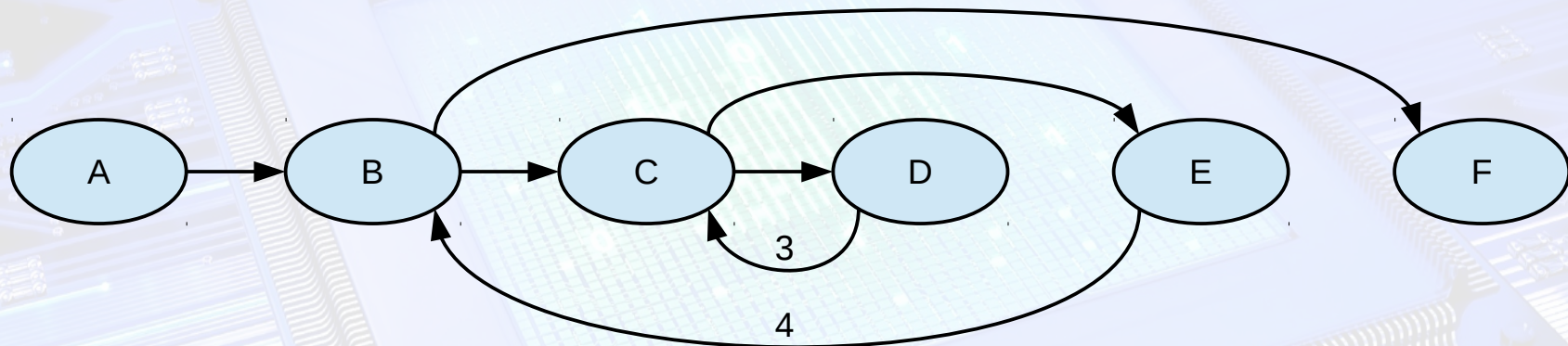
A B CDCDCDC EB CDCDCDC EB CDCDCDC EB CDCDCDC EB F

$\leq W$ potential misses $\leq W$ potential misses

We merge small windows

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A BCD CDCDCEBCDCDCDCEB CDCDCDCEBCDCDCDCEB F

Putting all together

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A

BCDCDCDCEBCDCDCDCEB

CDCDCDCEBCDCDCDCEB

F

Putting all together

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A

BCDCDCDCEBCDCDCDCEB

CDCDCDCEBCDCDCDCEB

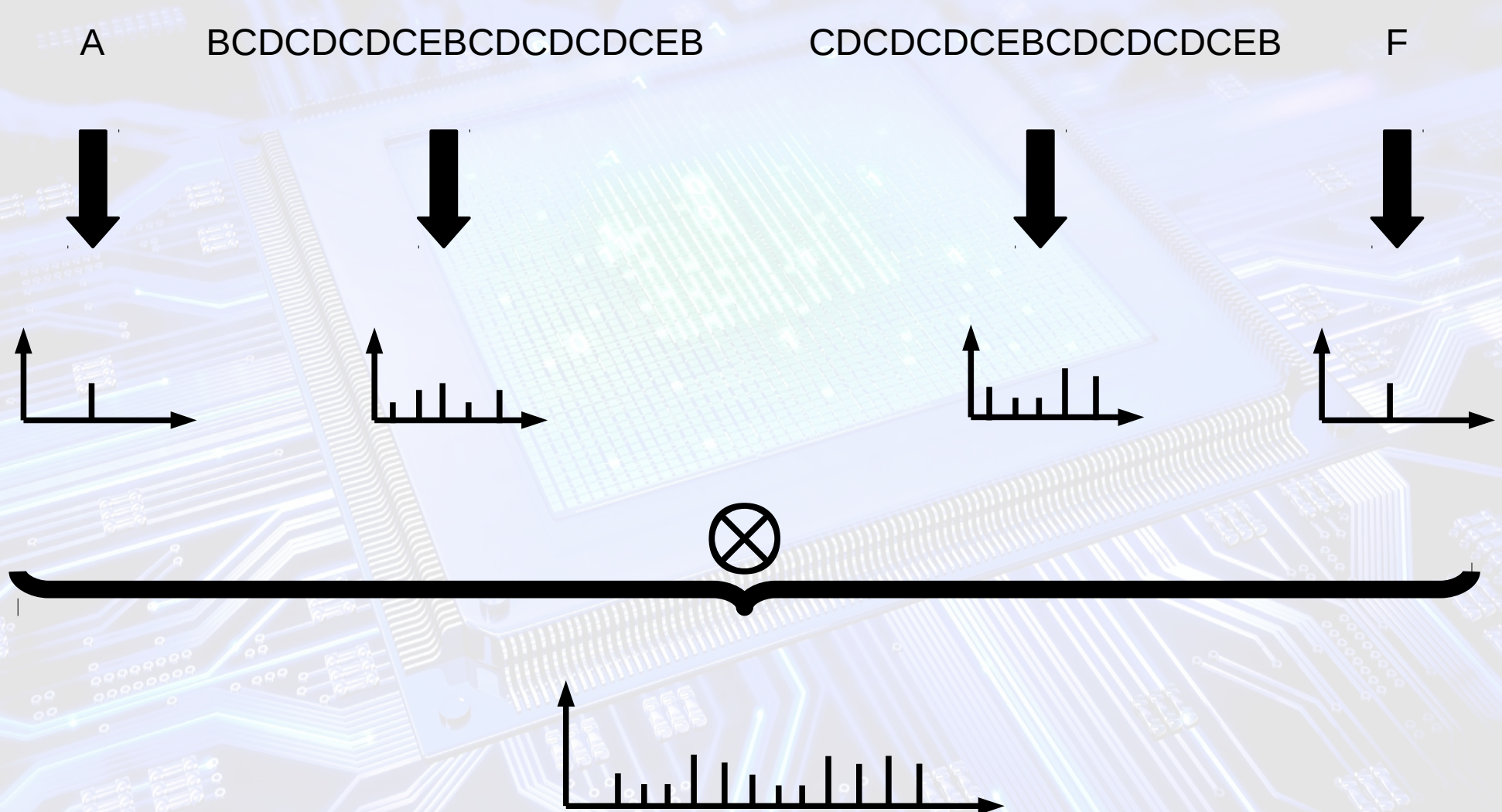
F



Putting all together

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Observations

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Precision:

- ▶ Windows share references $\Leftrightarrow$ Precision loss
- ▶ Infinite-size windows $\Rightarrow$ Exact distribution

Observations

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Precision:

- ▶ Windows share references $\Leftrightarrow$ Precision loss
- ▶ Infinite-size windows $\Rightarrow$ Exact distribution

Independence:

- ▶ Same initial state for all analyses of windows
- ▶ Convolution can be applied

Observations

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Precision:

- ▶ Windows share references $\Leftrightarrow$ Precision loss
- ▶ Infinite-size windows $\Rightarrow$ Exact distribution

Independence:

- ▶ Same initial state for all analyses of windows
- ▶ Convolution can be applied

Safety:

- ▶ Worst-case initial state
- ▶ Underestimation of hit probabilities
- ▶ Safe by construction

Evaluation of tightness

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Programs:

- ▶ Mälardalen benchmark suite
- ▶ 11 programs
- ▶ Single-path (or one trace if multi-paths)

Evaluation of tightness

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Programs:

- ▶ Mälardalen benchmark suite
- ▶ 11 programs
- ▶ Single-path (or one trace if multi-paths)

Cache:

- ▶ Set-associative (generalization of fully-associative)
- ▶ 8 sets, 4 ways, 32B lines (1KB cache)
- ▶ Modulo placement, random replacement

Evaluation of tightness

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Programs:

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- ▶ 8 sets, 4 ways, 32B lines (1KB cache)
- ▶ Modulo placement, random replacement

Approaches compared:

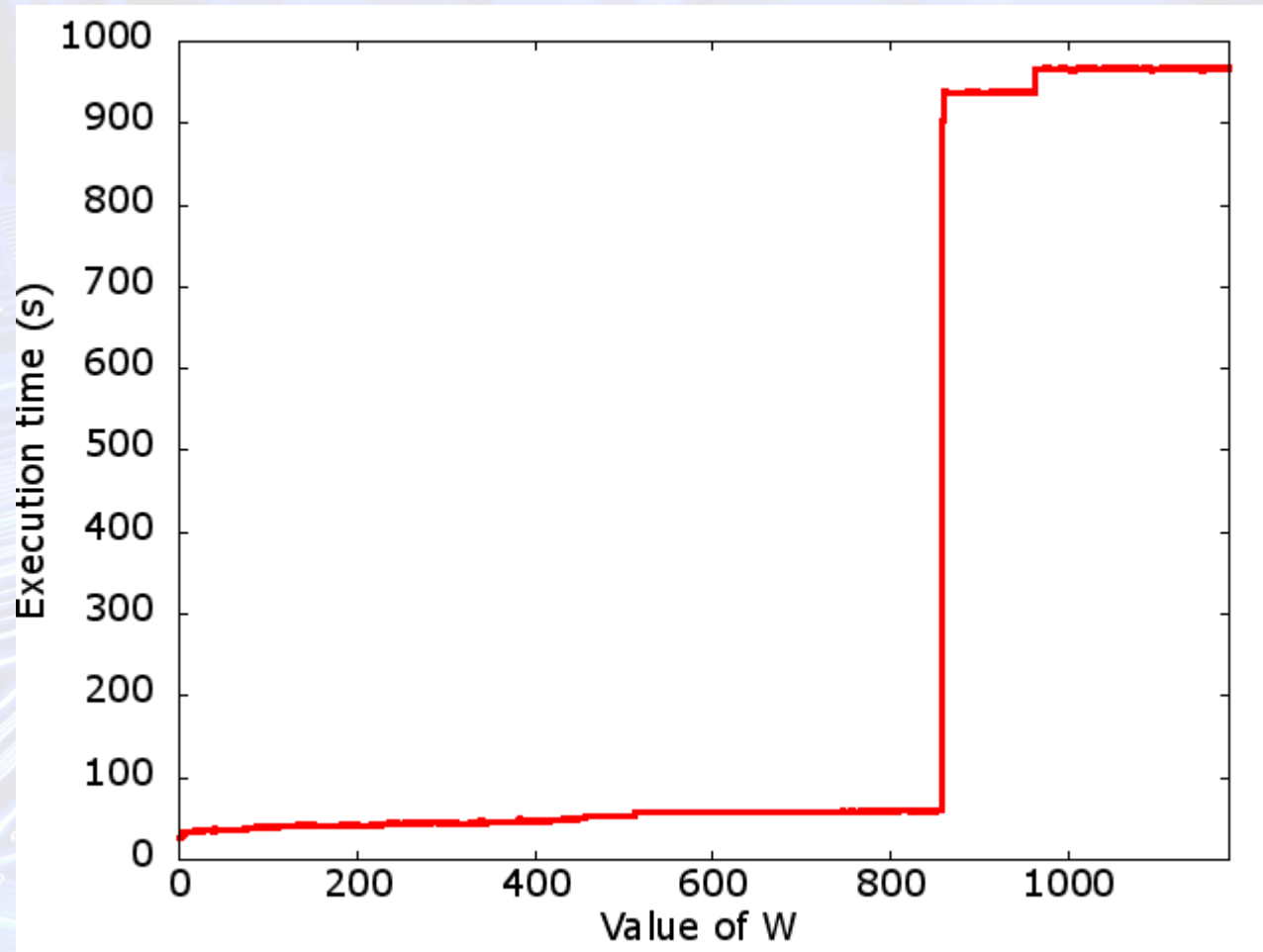
- ▶ Our approach (our implementation)
- ▶ Simulations (10000)
- ▶ Altmeyer & Davis [Alt2014] (~15mn max, always higher)

[Alt2014] S. Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis" (2014)

Evolution of execution time

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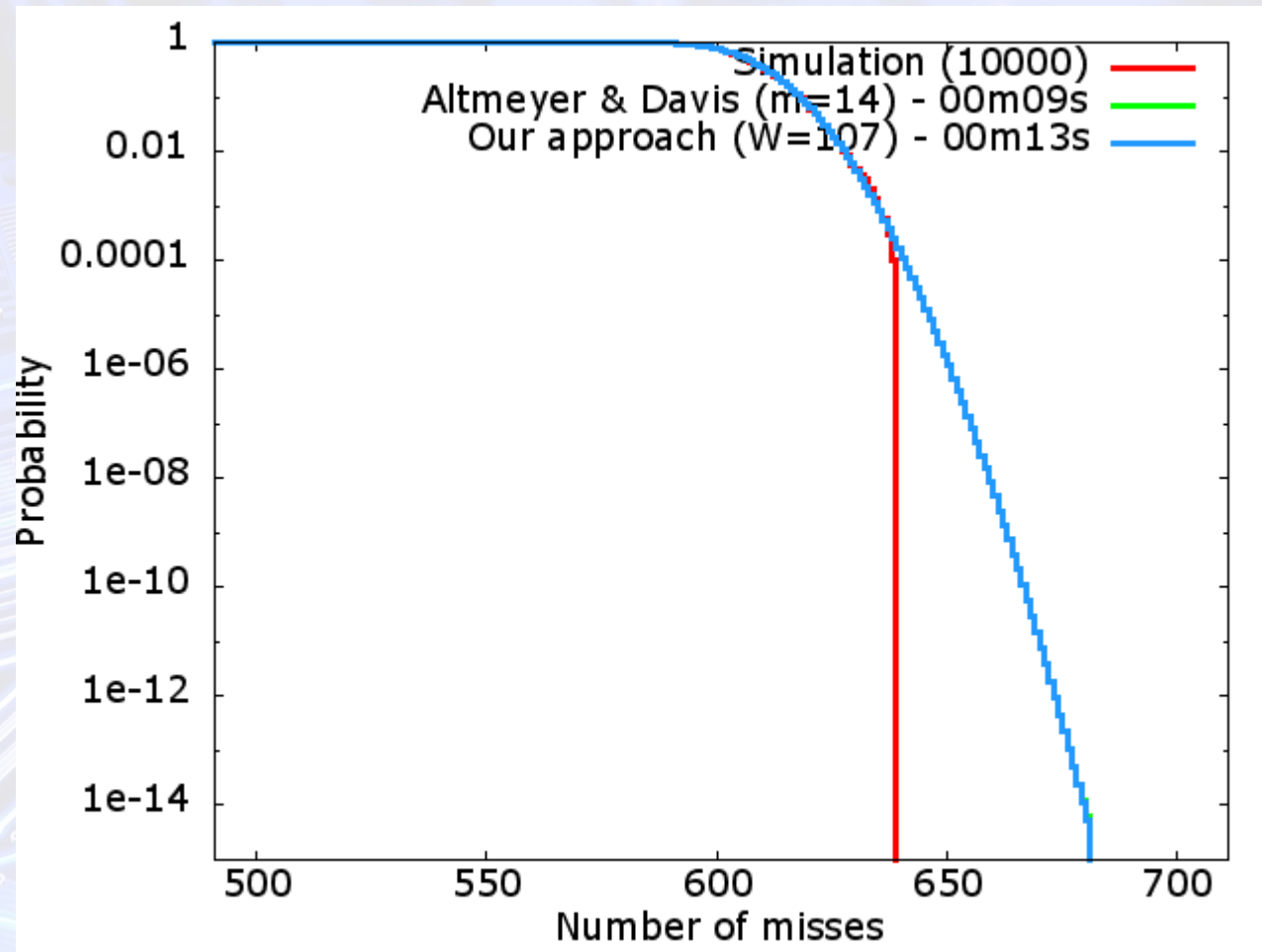


edn (39714 references)

Analysis of tightness for a small program

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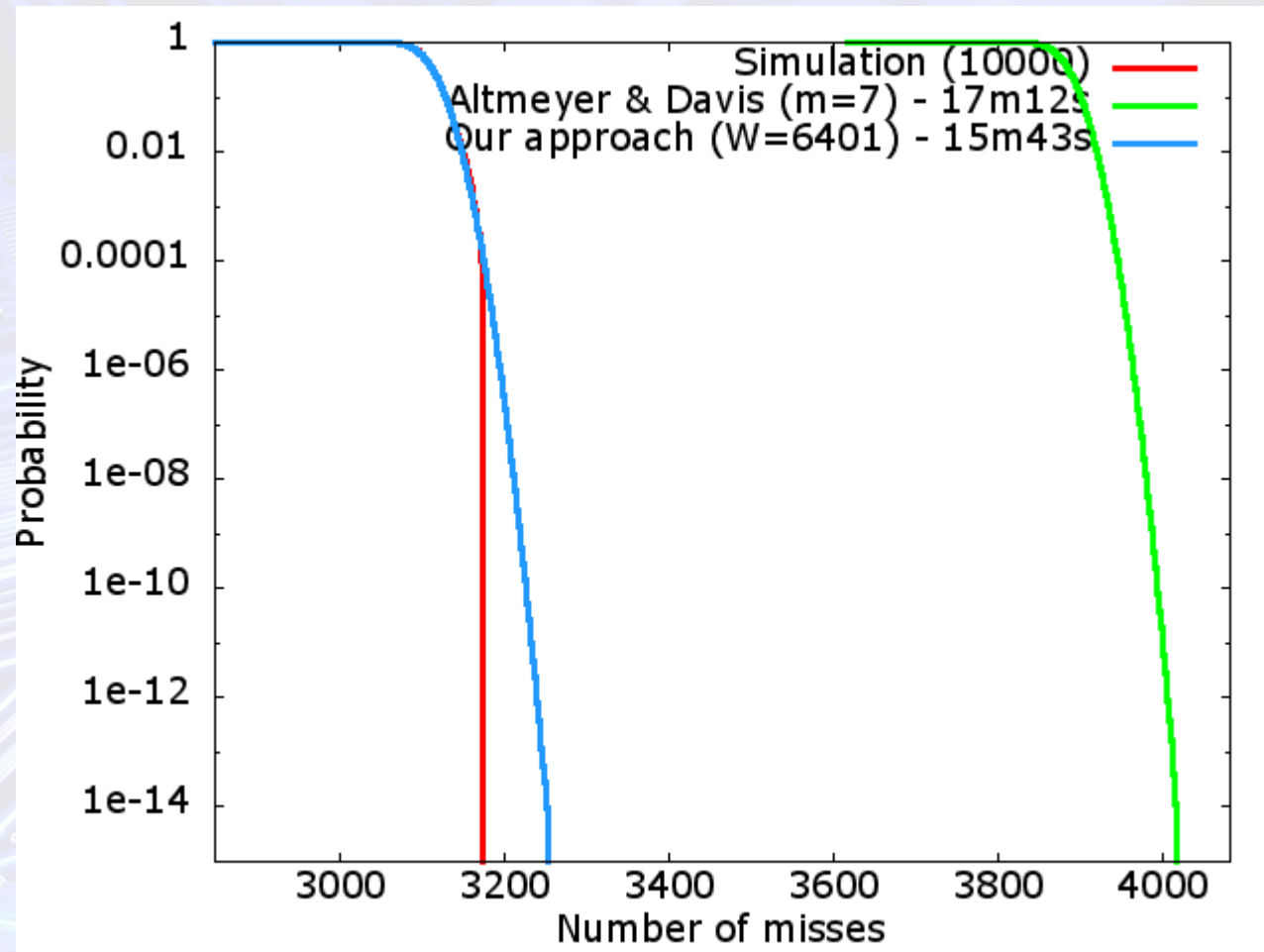


fdct (784 references)

Analysis of tightness for a medium-size program

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Reduction of the problem
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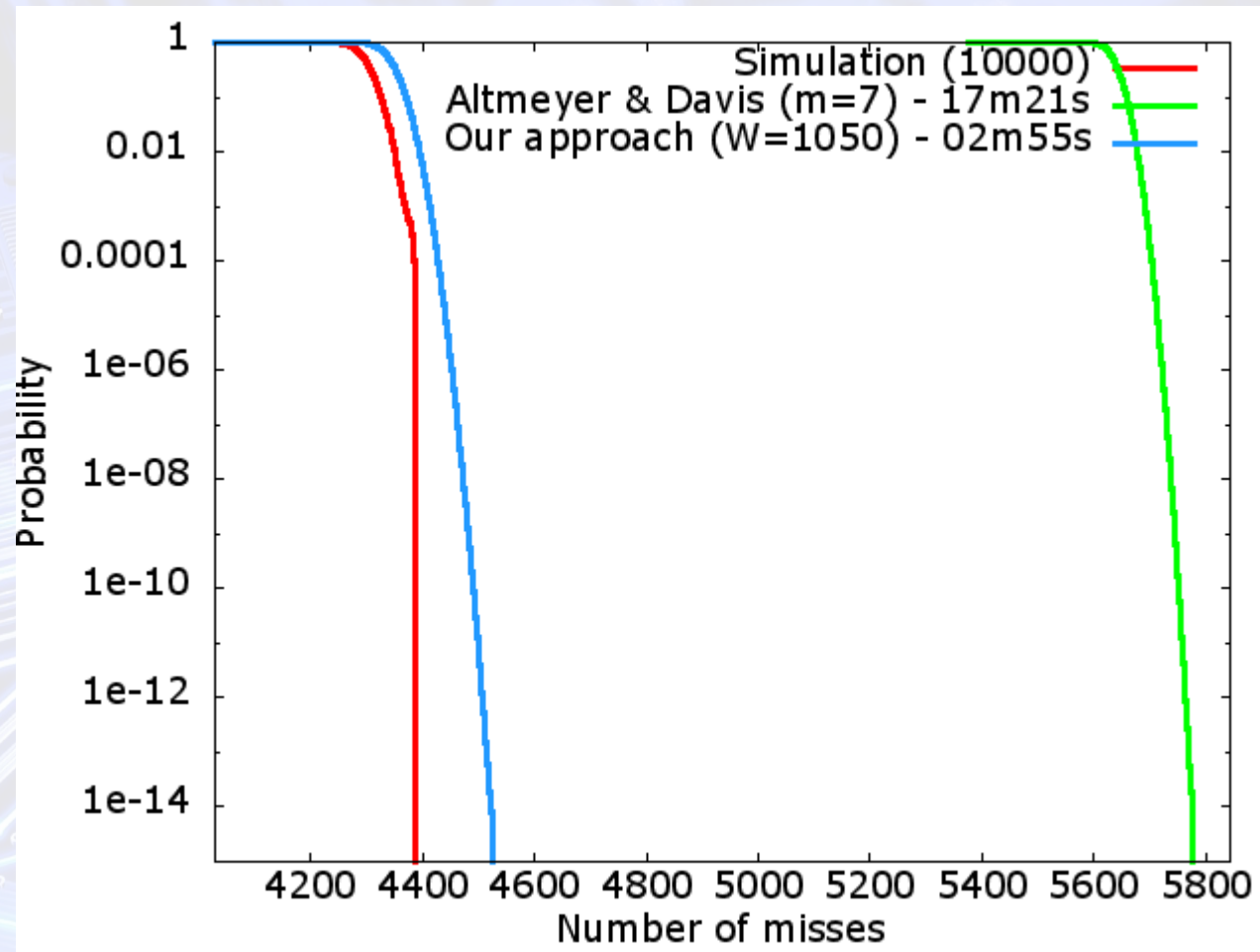


edn (39714 references)

Analysis of tightness for a bigger program

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Required background
Proposed method
Conclusion

Context
Exhaustive enumeration
Reduction of the problem
Experimental evaluation



fdct-jfdctint-minver-edn (83222 references)

Contribution

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Summary

Future work

References

Method:

- ▶ Static
- ▶ Probabilistic
- ▶ Random aspects of the cache

Contribution

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Future work

References

Method:

- ▶ Static
- ▶ Probabilistic
- ▶ Random aspects of the cache

Applicability:

- ▶ Random replacement caches
- ▶ Single-path programs

Contribution

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Future work
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Method:

- ▶ Static
- ▶ Probabilistic
- ▶ Random aspects of the cache

Applicability:

- ▶ Random replacement caches
- ▶ Single-path programs

Important properties:

- ▶ Safe by construction
- ▶ Tightness experimentally shown

Remaining work

Introduction

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Parallelization:

- ▶ Processing of windows
- ▶ Computation of tree

Remaining work

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Parallelization:

- ▶ Processing of windows
- ▶ Computation of tree

Combination of approaches:

- ▶ Partition into windows
- ▶ Altmeyer & Davis [Alt2014]

[Alt2014] S. Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis" (2014)

Remaining work

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Parallelization:

- ▶ Processing of windows
- ▶ Computation of tree

Combination of approaches:

- ▶ Partition into windows
- ▶ Altmeyer & Davis [Alt2014]

Extension to multi-path programs

[Alt2014] S. Altmeyer, R.I. Davis – "On the Correctness, Optimality and Precision of Static Probabilistic Timing Analysis" (2014)

Bibliography (1)

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- [Abe2014] Jaume Abella, Damien Hardy, Isabelle Puaut, Eduardo Quinones, and Francisco J. Carzorla. On the comparison of deterministic and probabilistic WCET estimation techniques. In 26th euromicro conference on real-time systems (ECRTS). To appear, 2014.
- [Alt2014] Sebastian Altmeyer and Robert I. Davis. On the correctness, optimality and precision of static probabilistic timing analysis. In Proceedings of the conference on design, automation & test in europe. European Design and Automation Association, 2014.
- [Ber2011] Dale Berger. A gentle introduction to resampling techniques. Claremont Graduate University.
- [Ber2002] Guillem Bernat, Antoine Colin, and Stefan M. Petters. WCET analysis of probabilistic hard real-time systems. In Proceedings of the 23rd IEEE real-time systems symposium. IEEE Computer Society, 2002.
- [Car2013] Francisco J. Cazorla, Eduardo Quiñones, Tullio Vardanega, Liliana Cucu, Benoit Triquet, Guillem Bernat, Emery Berger, Jaume Abella, Franck Wartel, Michael Houston, Luca Santinelli, Leonidas Kosmidis, Code Lo, and Dorin Maxim. Proartis: probabilistically analyzable real-time systems. ACM trans. embed. comput. Syst., 2013.
- [Col2001] Stuart Coles. An introduction to statistical modeling of extreme values. Springer, 2001.
- [Col2000] Antoine Colin and Isabelle Puaut. Worst case execution time analysis for a processor with branch prediction. RTS, 2000.
- [Cou1977] P. Cousot and R. Cousot. Abstract interpretation: a unified lattice model for static analysis of programs by construction or approximation of fixpoints. In Conference record of the fourth annual ACM SIGPLAN-SIGACT symposium on principles of programming languages. ACM, 1977.

Bibliography (2)

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- [Cuc2012] Liliana Cucu-Grosjean, Luca Santinelli, Michael Houston, Code Lo, Tullio Vardanega, Leonidas Kosmidis, Jaume Abella, Enrico Mezzetti, Eduard Quiñones, and Francisco J. Cazorla. Measurement-based probabilistic timing analysis for multi-path programs. In Robert Davis, 2012.
- [Dav2004] Laurent David and Isabelle Puaut. Static determination of probabilistic execution times. In Proceedings of the 16th euromicro conference on real-time systems. IEEE Computer Society, 2004.
- [Dav2013] Robert I Davis, Luca Santinelli, Sebastian Altmeyer, Claire Maiza, and Liliana Cucu-Grosjean. Analysis of probabilistic cache related pre-emption delays. In 25th euromicro conference on real-time systems (ECRTS). IEEE, 2013.
- [Eng2002] Jakob Engblom. Processor pipelines and static worst-case execution time analysis. PhD thesis. Uppsala University, 2002.
- [Fer1999] Christian Ferdinand and Reinhard Wilhelm. Efficient and precise cache behavior prediction for real-time systems. Real-time syst., 1999.
- [Har2011] Damien Hardy and Isabelle Puaut. WCET analysis of instruction cache hierarchies. J. syst. Archit., 2011.
- [Cos2013a] Leonidas Kosmidis, Jaume Abella, Eduardo Quiñones, and Francisco J. Cazorla. A cache design for probabilistically analysable real-time systems. In Proceedings of the conference on design, automation and test in europe. EDA Consortium, 2013.
- [Cos2013b] Leonidas Kosmidis, Luca Santinelli, Michael Houston, Liliana Cucu, Eduardo Quinones, Jaume Abella, Tullio Vardanega, and Francisco J Cazorla. A case study for a PTA-friendly processor. 2013.

Bibliography (3)

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- [Lea1983] MR Leadbetter. Extremes and local dependence in stationary sequences. Probability theory and related fields, 1983.
- [Max2012] Dorin Maxim, Mike Houston, Luca Santinelli, Guillem Bernat, Robert I. Davis, and Liliana Cucu-Grosjean. Re-sampling for statistical timing analysis of real-time systems. In Proceedings of the 20th international conference on real-time and network systems. ACM, 2012.
- [Max2013] Dorin Maxim. Analyse probabiliste des systèmes temps réel. PhD thesis. 2013.
- [Me12013] Alessandra Melani, Eric Noulard, and Luca Santinelli. Learning from probabilities: dependencies within real-time systems. In ETFA, 2013.
- [Pus1989] Peter Puschner and Ch. Koza. Calculating the maximum execution time of real-time programs. RTS, 1989.
- [Rei2007] Jan Reineke, Daniel Grund, Christoph Berg, and Reinhard Wilhelm. Timing predictability of cache replacement policies. Real-time syst., 2007.
- [Smi1982] Alan Jay Smith. Cache memories. ACM comput. Surv., 1982.
- [Ste1995] Yau-Tsun Steven Li and Sharad Malik. Performance analysis of embedded software using implicit path enumeration. In Proceedings of the 32nd annual ACM/IEEE design automation conference. ACM, 1995.
- [Wen2005] Ingomar Wenzel, Raimund Kirner, Peter Puschner, and Bernhard Rieder. Principles of timing anomalies in superscalar processors. In Proceedings of the fifth international conference on quality software. IEEE Computer Society, 2005.

Bibliography (4)

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- [Wil2008] Reinhard Wilhelm, Jakob Engblom, Andreas Ermedahl, Niklas Holsti, Stephan Thesing, David Whalley, Guillem Bernat, Christian Ferdinand, Reinhold Heckmann, Tulika Mitra, Frank Mueller, Isabelle Puaut, Peter Puschner, Jan Staschulat, and Per Stenström. The worst-case execution-time problem – overview of methods and survey of tools. ACM trans. embed. comput. Syst., 2008.
- [Wil2005] Nicky Williams, Bruno Marre, Patricia Mouy, and Muriel Roger. Pathcrawler: automatic generation of path tests by combining static and dynamic analysis. In In: proc. european dependable computing conference. volume 3463 of LNCS (2005) 281–292. Springer, 2005.
- [Zho2010] Shuchang Zhou. An efficient simulation algorithm for cache of random replacement policy. In, Network and parallel computing. Springer Berlin Heidelberg, 2010.

Static probabilistic timing analysis of worst-case execution time for random replacement caches

Student: Bastien Pасdeloup

Supervisors: Isabelle Puaut, Damien Hardy

Research team: ALF

Wednesday, June 25th 2014